Lectures

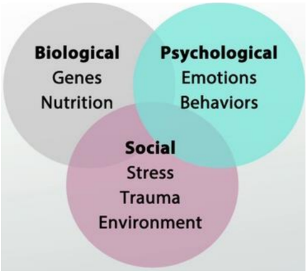
04/09/2023

L1 (PS 1,3)
Genes and Evolution

Psychology is:

- ***Mind*** refers to mental activity, which results from biological processes within the ***brain***.
- ***Behaviour*** = the description of the totality of observable human (or animal) actions.
- ***Psychologist*** = used broadly to describe someone whose career involves understanding mental life or predicting behaviour

<u>1) the genetic basis</u>

- Is it correct to claim that “a violent person has violent genes”?
  - In the 70’s
    - Environment → (violent) behaviour
  - In the 90’s
    - Genes → (violent) behaviour
  - Today
    - Genes ← → environment ← →(violent) behaviour
- Bio-psycho-social model
  - 

Source of Similarities and Differences

- Similarities such as showing similar emotions, following similar patterns in life
- Differences such as ability to learn maths, response to a stressful situation
- Sources of variability in mental processes and behaviours

The Genetic basis

- ***Genes*** = Functional biochemical units of heredity that make up the chromosomes
  - Meaningful sections of the DNA molecule
  - Govern the cell’s functioning by providing instructions for making proteins
  - Each gene is paired with another gene.
    - The pairs are located at corresponding positions on pairs of chromosomes
    - ***Allele*** is one specific variant of a gene.
- ***Genome*** = the blueprint that provides detailed instructions for every cell.
  - Whether a cell becomes part of the nose or bladder depends on which genes are turned on or off
    - This is determined by cues in- and outside of the cell
  - It contains all the genes in that organism

    - The human genome makes us human (and the genome for *drosophila* makes it a common house fly).
- Nearly every cell has 23 pairs of ***chromosomes*** (half from dad and ma)
    - situated in the nucleus of a cell

  v
- Made of ***deoxyribonucleic acid*** (DNA)
    - Consists of two intertwined strands of molecules

  v
- Segments of strands are genes (sequence of molecules along a DNA strand)

  v
- Each gene has specific instructions to manufacture a ***polypeptide***
    - The building blocks of protein, the basic chemicals that make up the structure of cells and direct their activities.
    - Environment determines which proteins are produced and when they are produced.

Heredity

- **Heredity =** the transmission of characteristics from parents to offspring by means of genes.
    - Heredity influences body weight, but there is no single "fatness gene".
        - Genes typically are not solo players.
- **Heritability =** the proportion of the variation in some specific traits in a population as a whole, not individual.
- Type of gene
    - A ***dominant gene*** from either parent is expressed whenever it is present.
    - A ***recessive gene*** is expressed only when it is matched with a similar gene from the other parent.
- Genotype & Phenotype
    - ***Genotype*** = an organism's specific set of genes, is determined at the moment of conception and never changes
    - ***Phenotype*** = observable physical characteristics and is always changing
        - Genetics, or nature, is one of the two influence
        - Environment, or nurture, is the second influence
- Polygenic effects
    - ***Single-gene characteristics*** are determined by one gene each.
    - ***Polygenic characteristics*** display a range of variability, the trait is influenced by many genes
        - Skin type has a huge range of skin tones (phenotype), which show that human skin colour is not the end product of a single dominant/recessive gene pairing (genotype)
- Sexual reproduction
    - Differences in siblings is because each person has a specific combination of genes.
    - ***Zygote*** = fertilised egg, which contains the 23 chromosomes
    - 8 million combinations of the 23 chromosomes from any two parents.
- Genetic mutations: advantageous, disadvantageous, ro both?
- ***Mutations*** = errors in cell division
    - Can be adaptive or maladaptive

        - Whether mutations are adapted depend on:
            - Survivability (darker moths in industrialised cities)
            - Dominance or recessiveness of a gene

<u>2) evolution by natural selection</u>

- The genome is shaped by evolution over the years → ***Darwin's evolution theory***
    - Charles Darwin hypothesised that all modern organisms
        - Are descended from a small set of shared ancestors
        - Have merged over time through the process of evolution
- The key mechanism of evolution is natural selection.
    - Three conditions
        - There is variation among individuals of a populations
        - Individuals with a certain trait survive and reproduce at higher rates than others
        - The trait associated with this advantage is passed from parents to offspring
    - Specific traits will be better represented in the next generation
- Organisms differ in genotype and variations in genotype are passed from generation to generation
- What matters is the survival of genes, *not* the survival of individuals
- The evidence for modern evolutionary theory comes from many sources
    - The fossil record
    - The resemblance between genomes of various organisms
    - Pseudogenes
    - Distribution of species across the world
        - Continental islands versus oceanic islands
- Despite overwhelming evidence, many people remain sceptical about the theory of evolution.
- It does *not* follow that evolution
    - Somehow improves organisms
    - Is the survival of the 'fittest'
        - Can only lead to rigid behavioural patterns

<u>3) Genes and behaviour</u>

Behavioural genetics = the study of how genes and environment interact to influence psychological activity

- Science indicates that genes lay the groundwork for many human traits.
- Who we are is determined by how our genes are expressed in distinct environments

Behavioural genetics methods

- ***Twin studies***
    - compares similarities between different type of twins to determine the genetic basis of specific traits
- ***Adoption studies***
    - compare the similarities between biological relatives and adoptive relatives

Nature and Nurture

- The Nature (genes) vs Nurture (environment) debate has become increasingly irrelevant.
    - Instead, there is a continuous interaction between genes and environment!
- ***Gene-Environment Interaction***
    - Some human traits are fixed
    - However, most psychological traits are liable to change with environmental experience.
    - Genes and environment affect our traits individually, but more important are their *interactive effects*

Epigenetics
- ***Epigenetics*** = a new field of genetic study, which looks at the processes by which environment affects ***gene expression***.
- Various environmental exposures do not alter DNA, but they do alter gene expression.
- Changes in how DNA is expressed can be passed along to future generations
    - ***Gene expression*** = Whether a gene is turned "on" or "off"
        - The extent to which a gene is transcribed into a sequence of amino acids (protein)
        - In each cell, some genes are expressed at any point in time and others are not.
        - This is controlled by the biochemical environment inside the cell.
            - The biochemical environment inside the cell is influenced by:
                - The environment outside the cell
                - Timing in development
                - The overall environment
                - Experience
                - Behaviour

Genetic modifications
- Researchers can employ various gene manipulation techniques to enhance or reduce the expression of a particular gene or even to insert a gene from one animal species into the embryo of another
- Changing one gene's expression leads to the expression or nonexpression of a series of other genes
    - Genes seldom work in isolation to influence mind and behaviour.
    - Rather, interaction among thousands of genes gives rise to the complexity of human experience

Optogenetics
- ***Optogenetics*** = a research technique, that provides precise control over when a neuron fires
    - Enables researchers to better understand the causal relationship between neural firing and behaviour.
        - E.g., turning off one set of neurons reduced cocaine use in animals addicted to that drug.
- A specific trait or behaviour is determined by the interaction between the environment (past and present) and

- One gene pair.
- Multiple gene pairs: polygenic inheritance.

Early stimulation is Critical

- Effects of prolonged deprivation
    - Species-universal experiences are required for fine-tuning neural connections
    - When expected experiences lack in *sensitive periods*, then the brain will fail to develop normally

Throwback

- Is it correct to claim that “a violent person has violent genes”?
- Genes ←→ environment ←→ (criminal) behaviour
- MAOA allele for low MAO activity ←→ severe maltreatment ←→ higher probability of being convicted of violent crimes

Cultural Influences

- Behaviour cannot be understood outside of the context in which it occurs
- ***Culture*** = shared way of life of a group of people
    - It is composed of behaviours, ideas, attitudes, values and traditions shared by a group.

Variation across culture

- Each culture develops ***Norms***
    - = rules for accepted and expected behaviour
- individualism-Collectivism Model is used commonly to explain cross-cultural differences

A word of caution

- There is no culture that is homogeneously individualistic or collectivistic
- Individualism/collectivism also varies within the culture

07/09/2023

L2 (PS 3)

The brain

- We’re able to predict behaviour based on brain scans
    - Like predicting criminality or discrimination

Studying the brain

- 19th century → Phrenology
    - bumps on the skull were interpreted in terms of personality traits
        - How intelligent you are, etc.
        - Used to proof white race was ‘the highest race’
- The brain is not that dissimilar to Phrenology

Phineas Gage

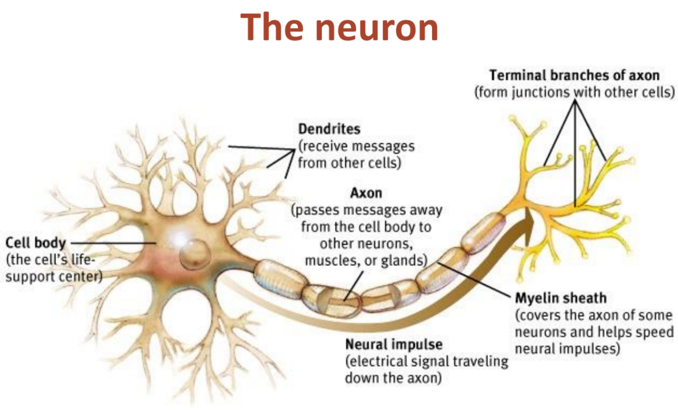
- Got a giant iron cylinder through his frontal cortex (impulse control) and got Post Traumatic Personality Change

    - Was very work persistent and after was completely unable to work and unable to hold a job due to aggression.
- His case led to a complete different perception to how we view the brain

Neurons
- There are certain areas in the brain that (neurons) that fire up when there is a specific shape in a specific part of your view that fires up
    - Lines, also houses, also faces

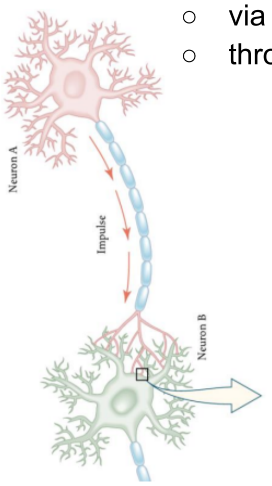
Methods for studying the brain
- Clinical observation of patients with brain damage
- Experimental techniques
    - Invasive: animal studies (e.g., lesions, single-cell recordings)
    - With TMS we can disrupt regions in the brain with no consequences (big ass magnet)
- Other techniques
    - Electrophysiology
        - EEG places electrodes and amplifies the electrical activity in the brain 30x and get reactions in milliseconds
        - You get a lot of lines, if you're only interested in a certain activity, you can turn it via a computer in an Event-Related Potential Technique (ERP)
    - Brain imaging
        - (f)MRI
        - PET scan (it's invasive and isn't really used in research)

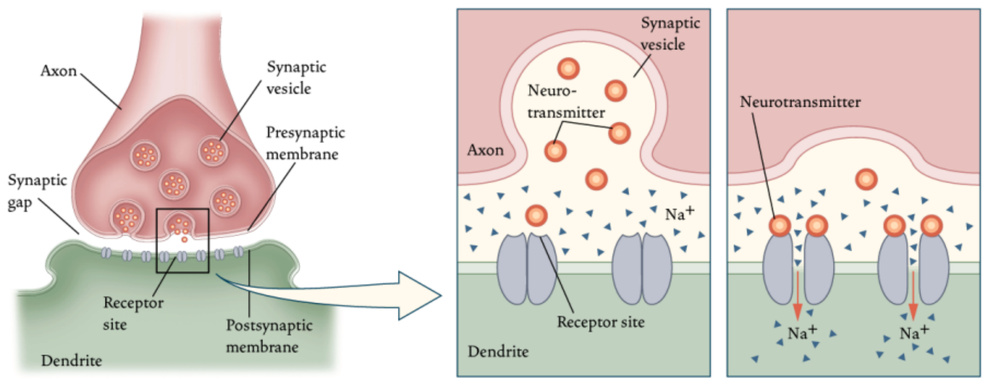
The nervous system
- The nervous system is made up of two basic kinds of cells
    - Glia
    - ***Neurons***, types:
        - Sensory receptors
        - ***Sensory neurons*** (afferent) **=** detect information from the physical world and pass it to the brain
        - ***Motor neurons*** (efferent) = direct muscles to contract or relax → movement
        - ***Interneurons* =** communicate only with other neurons
            - Afferent = → brain
            - Efferent = ← brain
- The nervous system is divided into two systems
    - ***The central nervous system*** (CNS)
        - Brain
        - Spinal cord



            - Evaluates and organises information received from the peripheral nervous system → directs PNS to perform behaviours or bodily adjustments.
    - ***The peripheral nervous system*** (PNS)
        - All other nerve cells in the rest of the body
            - Nerve cells = Neurons
        - Transmits variety of information to the central nervous system
- ***Autonomic nervous system (ANS)***: made up of two systems
    1. ***sympathetic nervous system***:
        - Arouses the body
            - Mobilising its energy in stressful situations
            - Uses energy
        - In defensive situations
            - The heart rate increases
            - The lungs expand to hold more oxygen
            - The pupils dilate
            - Blood flows to the muscles
    2. ***Parasympathetic nervous system***:
        - Calms the body
            - Conserving its energy
        - Bodily functions that occur in normal, non-stressful situations
        - After eating
            - The digestive process begins:
            - Nutrients are taken from the food and stored in the body

Communication among Neurons
- ***Action Potential***: is a **neural impulse**, a brief electrical charge that travels down an axon
    - very fast (3 km/h - 320km/h)
    - Three stages of electrical impulses of communications between neurons
        1. Reception phase: take chemical signals from other neurons
        2. Integration: assessing the incoming signals
        3. Transmission: pass their own signals to yet other receiving neurons.
- Neurons either fire or do not fire
    - All-or-none law
        - Intensity variations by
            - variations in the number of neurons firing (number).
            - variations in firing rate (frequency)
- Neurons interact
    - via synapses.
    - through chemical substances.

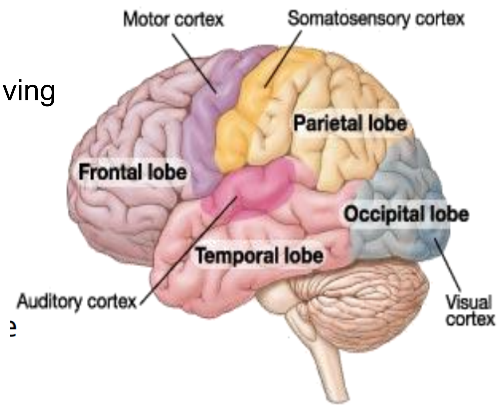




- ***Synapse***
    - the place where a signal passes from one nerve cell to another
- ***Neurotransmitters***
    - Chemical substances that transmit signals from one neuron to another.
        - Lock-and-key Model
        - Effect is terminated by
            - Autoreceptors
            - synaptic reuptake
            - Enzyme deactivation
        - The binding of a neurotransmitter with a receptor produces an excitatory (→) or inhibitory signal (<-).
- Drugs
    - ***Agonists***
        - increase of precursor
        - counteracting the cleanup enzymes
        - blocking the re-uptake
        - mimicking the neurotransmitter's action
    - ***Antagonists***
        - decrease precursor (or neurotransmitter)
        - increase effectiveness cleanup enzymes
        - enhance the re-uptake
        - blocking of receptors

The brain
- Visual cortex
    - The fMRI scan shows the visual cortex is active as the subject looks at faces.
    - Occipital lobe receives information from eyes and visual cortex gets activated

Lobes of the brain and problems
- Frontal lobe - personality, planning, emotion, problem solving
    - Motor cortex - movement
    - Broca's area - speech production
- Parietal lobe - touch
- temporal lobe - hearing
    - Wernicke's area - language comprehension
- Occipital lobe - vision



Brain Re-organization
- ***Plasticity*** = the brain's capacity for modification, as evident in brain reorganisation following
    - *"The brain is sculpted by our genes but also by our experiences."*

Brain
- Left hemisphere processes:
    - Language
    - Logic
    - Complex motor behaviour
- Right Hemisphere

    - Spatial ability: matching 3D image with 2D
    - Emotion: perceiving facial expressions and mood
    - Musical ability: perception of melodies
    - Some memory tasks: Learning tasks where context doesn't matter.

Summary
- Brain, the most complex structure of the human body, is the master of all neural processes that take place in the body.
- Brain is capable of reorganising itself in response to damage.
- Each hemisphere makes unique contributions to the integrated functioning of the brain.

11/09/2023

L3 (PS 4)
Consciousness

History of Consciousness
- Psychology began as a science of consciousness
- Behaviourists argued about alienating consciousness from psychology
- However, after 1960, mental concepts (consciousness)

Consciousness
- Subjective awareness of ourselves and our environment, resulting from brain activity.
- An awareness of the sensations, thoughts, and feelings that one is attending to at a given moment
- more controlled/focused when learning a complex concept or behaviour
    - Becomes automatic by practice
- Each of us experiences consciousness personally
    - We cannot know if two people experience the world in exactly the same way: explanation gap
- No specific part of the brain result in consciousness

Qualia
- ***Qualia***
    - the subjective or qualitative properties of experiences.
    - What it feels like, experientially, to see a red rose is different from what it feels like to see a yellow rose.

Consciousness arises as a function of which brain circuits are active
→ Global Workspace Model
- Prefrontal cortex: "I understand plans."
- Frontal motor cortex: "I'm all about movement."
- Parietal lobe: "I'm aware of space."
- Temporal lobe: "I hear things."
- Occipital lobe: "I see things"

Hypothesis: Specific patterns of brain activity can predict a person's behaviour:
- Conclusion: Type of awareness is related to which brain region processes the particular sensory information

Cognitive neuroscience

- If you think about throwing a basketball, the areas of the brain involved in the planning are activated.
    - Responsiveness in comatose patients

Readiness potential:

1 Readiness potential	2 Awareness of decision	3 Action
—---	---	-----------------→
-535 msec	-204 msec	0 msec

Variations in consciousness

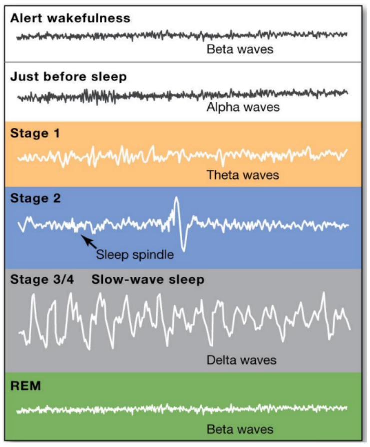
- Attention
- Sleep
- Drugs

Attention

- Attention refers to the process that enables you to focus selectively on some things and avoid focusing on others
- Rapid toggling between activities is the enemy of sustained, focused attention!
- But:
    - Cocktail party phenomenon (Cherry, 1953)
        - Sometimes the unattended information breaks through
        - The ability to attend selectively to only one voice among many.
- Unattended information may sometimes affect behaviour → subliminal perception
- Yet, without much awareness → ***change blindness***
    - A failure to notice large changes in one's environment.

- REM sleep is called the paradoxical sleep because you are paralysed but your mind is showing as much activity as you're awake

Sleep



- Conscious experience is largely turned off, to a certain extent it still processes information (analysing potential dangers, movement, etc.)
- Stages of sleep:
    - Alert wakefulness
        - Beta waves
    - Just before sleep
        - Alpha waves
    - Stage 1
        - Theta waves
        - Easily aroused/awakened, or not even realise you are sleeping, Falling sensation/fantastical images
    - Stage 2
        - still theta-waves with ***sleep spindles***
            - occasional bursts of activity
        - ***K-complex***
            - occasional large waves

- can be triggered by abrupt noises and may play a role in staying asleep
    - Stage 3 / 4
        - ***delta waves (slow-wave-sleep)***
            - large, regular brain patterns
            - Still processing information about the environment (eg: potential danger).
        - hard to wake, disoriented when they do.
    - Stage 5
        - Beta waves
            - Beta wave activity represents awake, alert mind, but paralyses muscles (sleeping body with active brain)
        - After about 90 minutes, the cycle returns to stage 1, with so-called ***REM-sleep phases (rapid eye movement)***.
            - When awakened during REM-sleep, people often report dreaming.

Theories about sleep

- ***Restorative theory***
    - Allows the body to rest and repair itself
    - Clears out toxic metabolic by-products.
    - Long sleep deprivation causes
        - mood problems
        - decreased cognitive performance
        - reduced short-term memory
        - compromises immune system
        - death through microsleeps (short sleep phases during day-time-activity)
        - increased activation of serotonin receptors → sometimes alleviates depression.
- ***Circadian rhythm theor***y
    - Sleep has evolved to keep animals quiet during dangerous times of the day (adaptive).
    - Animal's typical amount of sleep depends on
        - how much time that animal needs to obtain food,
        - how easily it can hide
        - how vulnerable it is to attack.
    - Humans sleep at night because our great dependence on vision makes us vulnerable.
- ***Facilitation of learning theory***
    - neural connections made during the day strengthen at night, especially during dreaming about the knowledge and REM-sleep

Dreams

- = products of an altered state of consciousness in which images and fantasies are confused with reality.
    - Non-REM dreams: dull
        - general deactivation of many brain areas.

    - REM dreams: bizarre and intense
        - Some brain areas show increased activity.
    - REM is correlated with dreams, it doesn't cause them
    - ***Activation-synthesis hypothesis***:
        - a hypothesis of dreaming proposing that the brain tries to make sense of random brain activity that occurs during sleep by synthesising the activity with stored memories.

Drugs

- ***Tolerance***:
    - increasing amounts of a drug needed to achieve the intended effect
- ***Addiction***:
    - Drug use that remains compulsive despite its negative consequences
    - Physical and psychological dependence
- ***Withdrawal***:
    - Physiological and psychological state characterised by feelings of anxiety, tension, and cravings for the addictive substance
- ***Psychoactive drugs***
    - can be used to alter physical sensations, levels of consciousness, thoughts, moods, and behaviours in ways that users believe are desirable.
    - Change the brain's neurochemistry by manipulating neurotransmitter-systems.
        - Either by imitating the brain's natural neurotransmitters (eg: marijuana, opiates)
        - or changing the activity of particular neurotransmitter receptors (eg: cocaine).
    - ***Stimulants***
        - heighten the activity of the CNS (increase behavioural & mental activity) and the sympathetic nervous system (increasing heart rate and blood pressure)
        - *Amphetamines*
            - causes an increase of dopamine in synapses.
            - Reduces fatigue and can cause weight loss, but leads to insomnia, anxiety and potential for addiction.
        - *Methamphetamine*
            - breaks down to amphetamine in the body and has a long-lasting effect.
            - Produces very high levels of dopamine, but over time depletes dopamine levels and damages various brain structures.
            - Affects memory & emotion and causes physical damage in long-term users.
        - *Cocaine*
            - causes an increase of dopamine in synapses.
            - Gives a wave of confidence and energy.
            - Habitual use in large quantities can lead to paranoia, psychotic behaviour & violence.
        - *Nicotine*
        - *Caffeine*

    - ***Depressants:***
        - depress the activity of the CNS (reduce behavioural & mental activity)
        - *Alcohol*
        - *Anti-anxiety drugs - benzodiazepines*
        - *Barbiturates*
    - ***Opiates (narcotics):***
        - Depress or slow down the central nervous system, bind with endorphin-receptors; relief pain and provide pleasure, relaxation and euphoria.
        - Highly addictive because they increase pleasure by binding with mu opioid receptors and increase wanting of the drug by indirectly activating dopamine receptors.
        - *Heroin*
        - *Morphine*
        - *Codeine*
    - ***Hallucinogens (psychedelics):***
        - Produce alterations in cognition, mood, and perception
        - *LSD*
        - *Mescaline -from peyote cactus-*
        - *Psilocybin mushrooms*
- Many commonly used drugs do not fit into these four major categories (e.g., marijuana, MDMA)
- ***Endogenous attention***
    - Intentionally directing the focus of your attention in this way
- ***Exogenous attention***
    - When the focus of your attention is driven by a stimulus or event.
- ***Unconscious*** processes influence people's thoughts and actions as they go through their daily lives.
- ***Priming*** is a facilitation in the response to a stimulus due to recent experience with that stimulus or a related stimulus.
- ***Subliminal perception***
    - occurs when stimuli are processed by sensory systems
    - because of their short duration or subtlety, they do not reach consciousness
- ***Automatic processing***
    - When a task is so well learned that we can do it without much attention
- ***Controlled processing***
    - Difficult or unfamiliar tasks require people to pay attention
- Persons subjective consciousness varies within a day
    - ***Altered consciousness*** = a state that changes your subjective perception of consciousness from how you typically experience it

Meditation

- ***Meditation*** = a mental procedure that focuses attention on an external object, an internal event, or a sense of awareness.
    - ***Concentrative meditation***
        - Focus attention on one thing, such as breathing pattern, mental image
    - ***Mindfulness meditation***

            - Let thoughts flow freely, paying attention to them but trying not to react to them

Hypnosis
- ***Hypnosis***
    - a social interaction during which a person responds to suggestions, experiences changes in memory, perception, and/or voluntary action.
- Psychologists generally agree that hypnosis affects some people,
    - but they do not agree on whether it produces a genuinely altered state of consciousness.
- ***Posthypnotic suggestion***: suggestion of the hypnotist, that after the hypnosis session, the listener will experience a change in memory, perception, or voluntary action.
    - Posthypnotic suggestions can subtly influence some types of behaviours.
- ***Hypnotic suggestibility***: some people can be hypnotised easier than others
    - related to personal qualities such as being focused, imaginative, easily absorbed, and willing to participate.
    - No evidence indicates that people will do things under hypnosis that they find immoral or otherwise objectionable.
- ***sociocognitive theory of hypnosis*** = hypnotised people behave as they expect hypnotised people to behave, even if those expectations are faulty.
- ***Neodissociation theory of hypnosis*** = theory that says; hypnosis is a trancelike state in which conscious awareness is separated, or dissociated, from other aspects of consciousness.
    - Numerous brain imaging studies have supported the neodissociation theory of hypnosis.
- ***Hypnotic analgesia***: a form of pain reduction by using hypnosis.
    - effective in dealing with immediate pain and chronic pain.
    - The patient feels the sensations associated with pain but feels detached from those sensations.
    - A patient can also be taught self-hypnosis to deal with pain.
- Individuals differ tremendously in the number of hours they sleep.
    - Gene ***sleepless*** regulates a protein that reduces action potentials in the brain
    - Loss of that gene is loss of sleep

Circadian rhythm
- = biological pattern that regulates brain activity and other physiological processes at regular intervals as a function of time of day.
- Many brain regions are more active during sleep than during wakefulness.

Sleeping brain
- Multiple brain regions are involved:
- Information about light is sent to a small region of the hypothalamus called the ***suprachiasmatic nucleus***,
  v
- This sends signals to the pineal gland.
  v

- Pineal gland sends the hormone ***melatonin***, which travels through the bloodstream and affects various receptors, including the brain.
    - Bright light suppresses the production of melatonin, darkness triggers its release.
    - Melatonin is necessary for the circadian cycles that regulate sleep.
- During sleep, conscious experience is largely turned off, but to a certain extent it still processes information (analysing potential dangers, body movement, …).

Sleep disorders

- ***Insomnia***
    - difficulty with falling asleep and staying asleep
    - interferes with mental health and ability to function in daily life, often getting worse by worrying about sleep.
    - Taking sleeping pills may work in the short run but lead to dependency.
    - Successful treatment for insomnia combines drug therapy with cognitive-behavioural therapy
    - ***Pseudo-insomnia***
        - Dreaming you are not sleeping
- ***Obstructive sleep apnea***
    - Stopping breathing during sleep for short periods, through closing of the throat.
    - Struggle to breathe awakes the patient.
        - Awakenings often don't get remembered.
    - Main-symptom is loud snoring
    - Can cause
        - daytime fatigue
        - inability to concentrate
        - in long term strokes
- ***Narcolepsy***:
    - excessive sleepiness that lasts for several seconds to minutes.
    - During an episode they may experience the muscle paralysation of REM-sleep, causing them to collapse.
    - Can be a genetic condition or an autoimmune disorder.
- ***REM behaviour disorder***:
    - normal paralysis that accompanies REM sleep is disabled
    - act out dreams during REM-sleep
- ***Somnambulism***:
    - Sleepwalking
    - Glassy-eyed
    - Disconnected appearance
- ***Sleep paralysis***:
    - wake up during REM sleep muscle paralysation

Changes in brain structure

- ***Traumatic brain injury (TBI)***:
    - = caused by a blow or very sharp movement of the head resulting in impairment in mental functioning
        - impair thinking

        - Memories
        - emotions
        - personality
    - Can cause short term or long-term disabilities.
    - ***Concussion*** causes brain swelling, resulting in brain damage that can have long-lasting effects.
        - Signs of concussion include
            - mental confusion
            - Dizziness
            - a dazed look
            - memory problems
            - Sometimes the temporary loss of consciousness.
        - People usually recover in 1-2 weeks.
- ***Coma***:
    - surviving is the first step to recovery, a coma (as result of a TBI or medically induced) allows the brain to rest.
    - ***Minimally conscious state***:
        - Some people who emerge from a coma are able to make deliberate movements and may even try to communicate.
    - ***Unresponsive wakefulness syndrome***:
        - people who seem to have emerged from the coma, have sleep/wake cycles, open eyes, but are not able to respond
        - they have no consciousness.
    - Some people in comas are conscious of what is happening around them and may be able to communicate.
    - Measuring brain metabolism via PET imaging identifies which patients in unresponsive states are likely to regain consciousness.
- ***Brain death***:
    - = irreversible loss of brain function
    - no activity in any region of the brain
    - The body quickly stops functioning.
- (partly) ***Locked-in syndrome***:
    - (nearly) all muscles are paralyzed,
    - no way of communication
    - still total awareness (sometimes little communication through vertical eye-movement).

14/09/2023

L4 (PS 5)

Sensation & Perception

Sensation is not perception
- The underside of our brain’s right hemisphere → helps us recognize a familiar human face as soon as we detect it
    - Prosopagnozia example
- Human ears are most sensitive to sound frequencies that include human voices, especially a baby’s cry
- Frogs, which feed on flying insects, have cells in their eyes that fire only in response to small, dark, moving objects

The senses
- ***Sensation*** - the detection of physical stimuli and the transmission of this information to the brain
- ***Perception*** - thye processing, organisation, and interpretation of sensory information and results in our conscious experience of the world
- Sensation and perception are integrated into experience and experience guides sensation and perception. (two-way street)
- Distal stimulus → proximal stimulus → Transduction → Sensation → Perception
    - ***Transduction***
        - sensory systems translate physical properties of stimuli into patterns of neural impulses.
        - features of the physical environment are coded by activity in different neurons.

    v
    - Takes place at ***sensory receptors*** = specialised cells within each sense organ that send signals from our environment to activate neurons.
        - They receive ***physical stimulation*** (vision, hearing, touch) or ***chemical stimulation*** (taste, smell).

    v
    - Neural impulses get passed to the brain (first to the **thalamus** and then to the **cerebral cortex**, except smell goes directly to cortex).
        - Each sense organ responds to specific stimuli and contains specific receptors and pathways for each sense.

Information
- ***Qualitative information***
    - Consists of basic qualities of stimulus (difference between salty and sweet).
    - Different sensory receptors respond to qualitatively different stimuli.
- ***Quantitative information***
    - The degree/magnitude of those qualities (relative saltiness or sweetness).
        - Different quantities are coded by the frequency in which the neurons fire

Psychophysics
- Examine our psychological experiences of physical stimuli and the relationship between psychological sensory experience and the physical characteristics of a stimuli

Thresholds
- We have very restricted awareness:
    - ***Absolute threshold***
        - The minimum intensity that must occur before you experience a sensation.
        - It is the stimulus intensity you would detect 50% of the time.
    - ***Subliminal***
        - Below absolute threshold
        - If you cannot detect it *consciously* at least 50% of the time, it is

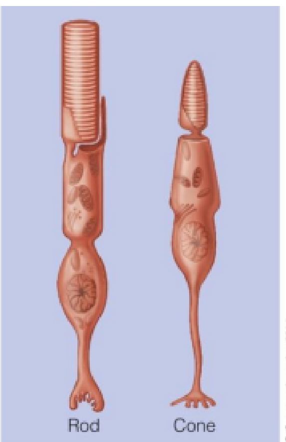
        subliminal for you
    - ***Difference threshold***
        - The smallest difference between two stimuli that we can notice 50% of the time.
        - It is based on the proportion of the original stimulus, rather than on a fixed amount of difference.
            - The more intense the stimulus, the bigger the change needed for you to notice: ***Weber's law***.
        - We experience the difference threshold as a just noticeable difference
- The ***signal detection theory*** = detecting a stimulus is not an objective process. It is a subjective judgement with two components:
    - the sensitivity to the stimulus in the presence of distraction of other stimuli
    - the criteria used to make the judgement from ambiguous information
        - The judgement can be influenced by knowledge about the situation and awareness of consequences
    - There are 4 possible outcomes when a participant is asked whether a signal occurred during a trial:
        - ***hit***: signal is presented and gets detected
        - ***miss***: signal is presented but not detected
        - ***false alarm***: signal was not presented but gets "detected"
        - ***correct rejection***: signal is not presented and doesn't get detected
    - Computing the hit-rate with the false alarm-rate can correct any biases.
    - ***Response bias*** is a participant's tendency to report or not report detecting the signal in an ambiguous trial.

The senses
- Types of senses:
    - Vision
    - Hearing
    - Taste
    - Smell
    - Skin senses (pressure, temperature, pain)
    - Vestibular sense
    - Kinesthesia
    - Proprioception
    - Interoception
- ***Quantitative variations*** (intensity)
    - Rates of neuron firing
    - Total number of neurons triggered
- ***Qualitative variations*** (sensory quality)
    - Different sensory qualities are signalled by different neurons
    - Certain sensory qualities arise because of different patterns of activation across a whole set of neurons
- Sensory adaptation
    - A decrease in sensitivity to a constant level of a stimulus
    - When the continuous stimulus stops, the sensor system usually responds to the absence of it.
    - Benefit:

        - Freedom to focus on informative changes in our environment
- Perception results from a construction of signals into a stable, unified representation of the world, most things perceived involve more than one sensation.
    - representation is not always perfect, sometimes this representation can have unusual results: ***Synesthesia***
        - = unusual experience where, for example, a visual image has a taste or a sound. One area of the brain might have adopted another area's role.

Vision - the retina

- ***Visual perception*** is our most important source of knowledge; it allows us to perceive information from distance.
- Very little of seeing takes place in the eyes, vision results from constructive processes that occur throughout the brain (damage in the visual cortex will impair vision, even if the eyes are normal)
- Light (direct or reflected)
    - Can vary in intensity and wavelength
    - Is absorbed by pigments in the ***photoreceptors*** (rods and cones), light-sensitive chemicals initiate the transduction of light waves to electrical neural impulses.
        - ***Photopigments*** (contained by the photoreceptors) are protein molecules that split apart when exposed to light.
        - decomposition of those photopigments alter the membrane potential and trigger action potentials in downstream neurons.
    - other cells (in the middle layer of the retina) compute information & converge the output to the retinal ***ganglion cell***.
        - First neurons to actually generate action potentials and axons
    - Those axons bundled into the ***optic nerve*** (blind spot)
        - Send the signals to the thalamus
- Light → Cornea → Pupil, Iris → Lens → Retina → Optic nerve → Brain’s visual cortex



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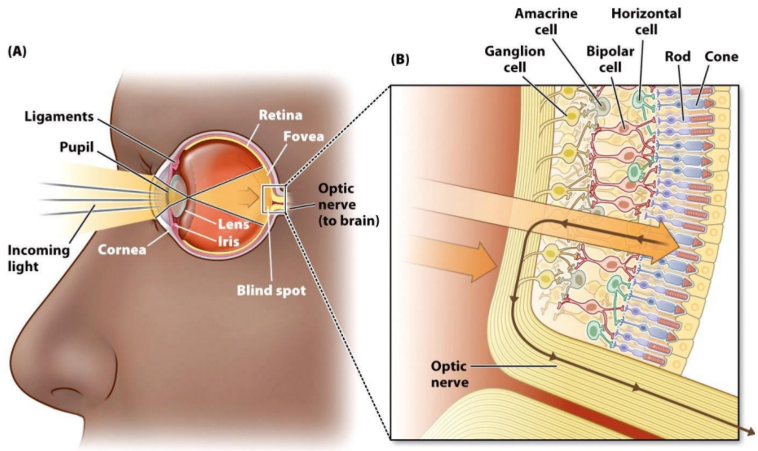
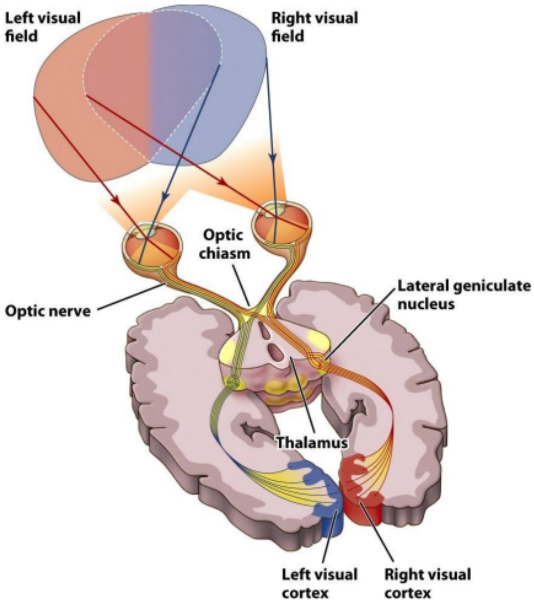


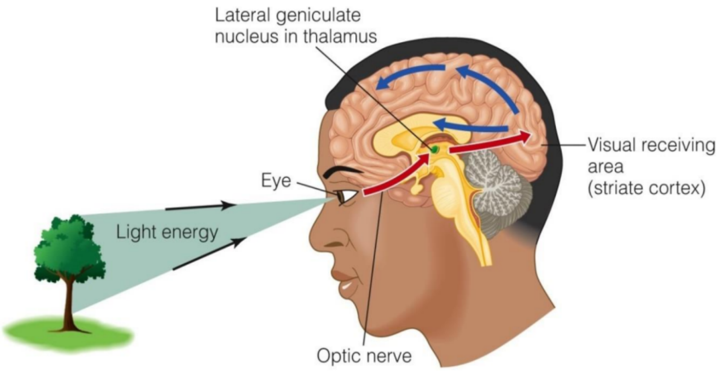
- ***Rods*** → night vision, no colour vision
    - react to extremely low levels of light, no colour vision/fine details, responsible for night vision
- ***Cones*** → day vision, colour vision

        - less sensitive to low levels of light, vision of colour & details, vision under brighter conditions
- ***Fovea***: small region near the retina's centre, densely packed with cones (no rods).
- Cones get rare to the edges of the retina, while rods are concentrated at the edges
- Colour vision
    - Colour is not the property of an object & doesn't exist in the physical world. It's a product of our visual system.
    - Light consists of electromagnetic waves, and the wavelength determines the colour we perceive.
    - different theories how cone cells transduce light into neural impulses:
        - Trichromatic theory (Young-Helmholz)
            - colour vision results from activity in three different types of cones, sensitive to different wavelengths
            - 1 type is each sensitive to *short*(blue-violet), *medium*(yellow-green) & *long*(red-orange) wavelengths.
            - BUT: trichromatic theory doesn't explain why red-green colour-blind people can see yellow...
        - Opponent-process theory (Hering)
            - certain colours are opponents (red-green; blue-yellow, …) If we stare at a red picture the receptors for red become fatigued, while those for green don't and the afterimage appears to be green.
            - Ganglion cells working in opposing pairs create the perception that certain colours are opponents.
            - There is no such thing as "blue-yellow" or "red-green"
            - Black-white
            - Red-green
            - Blue-yellow
        - ^ both represent different stages of visual processing
    - Colour has 3 dimensions:
        - ***hue***: the distinctive characteristics that place a particular colour in the spectrum (this depends primarily on the lights dominant wavelength)
        - ***saturation***: the purity of the colour varies according to the mixture of the wavelength in a stimulus, more different wavelengths give a less pure colour and vice versa.
        - ***lightness***: the colour's perceived intensity, mostly determined by the total amount of light reaching the eye, but also by the background it appears on.
    - The visual system uses a number of organising principles to help determine the meaning of visual input or the most likely interpretation based on experience with the world
    - We automatically divide visual scenes into objects and background (***figure and ground***)

Vision - beyond the retina

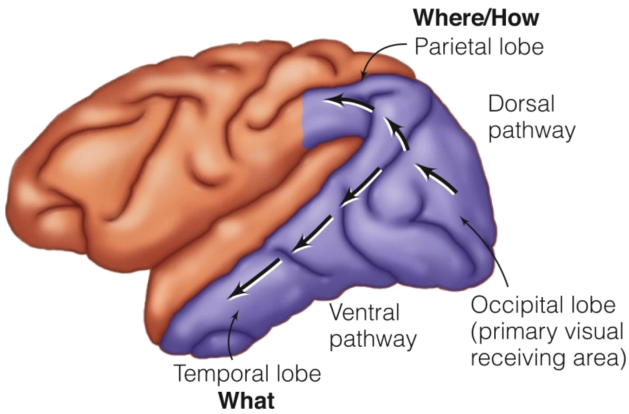
- What happens after the retina?
    - Picture →





- ***Optic chiasm***
    - Half of the optic nerves cross.
    - Information from one side of the visual field goes to the opposing hemisphere.
- The information reaches the visual area of the thalamus, then travels to the ***primary visual cortex.***

- Two parallel processing streams/pathways:
    - *"Where"* upper dorsal stream(path to parietal cortex)
        - specialised for spatial perception (determining where an object is and relating it to other objects in the scene)
        -
    - *"What"* lower ventral stream (path to temporal cortex)
        - specialised for the perception and recognition of objects (determining their colour and shape)
    - ***Object agnosia*** is the inability to recognize objects visually, but they can be identified by touch or sound. And can be reached for because the "where" stream is intact.
    -
- There is without a doubt a strong modularity within each system



Visual perception
- The visual system is not a passive receiver of visual information but modulates, organises, and interprets this information!
- Perception occurs on the basis of
    - Bottom-up (data-driven) processes

        - is based on the physical features of the stimulus, because of our present experience of the sensations.
        - Starts at your sensory receptors and works up to higher levels of processing.
    - Top-down (knowledge-driven) processes
        - is based on knowledge, expectations and past experiences, they shape the interpretation of the sensory information.
        - Context affects perception: what we expect to see (higher level) influences what we perceive (lower level).
        - E.g., priming
        - E.g., context
        - BUT: the orientation of an individual part is sometimes crucial for the perception of the whole.
        - Constructs perceptions from this sensory input by drawing on your experience and expectations.

Context

- Our immediate context, and the motivation and emotion we bring to a situation, also affect our interpretations.
- People's expectations influence their perceptions constantly, whether about an angry man
    - Is he reaching for his keys or a weapon?

What determines our perceptual set?

- Emotions can shove our perceptions in one direction or another
    - Hearing sad music can predispose people to perceive a sad meaning in spoken homophonic word
        - Mourning - morning
        - Die - dye
        - Pain - pane
- Experience → concepts, schemas
    - Helps organise and interpret unfamiliar information
    - Apply top-down processing to interpret ambiguous sensations
- Perception is influenced by culture.

Perceptual organisation

- Segregation of figure and ground
    - Principles of figure-ground segregation
        - Lower areas are more often perceived as figure than higher areas
        - Convex forms are more often perceived as figure whereas concave forms are more often perceived as background
- Perceptual interpretation on the basis of organising principles (***Gestalt theory:***)
    - perception is more than the result of accumulating sensory data, the brain uses innate principles to organise a collection of features (metal, tires, glass,…) into a whole unit (a car). Rules:
        - ***proximity***: the closer two figures are
        - ***similarity***: the more two figures resemble each other (in shape, colour, orientation), the more likely we are to group them as part of the

same object
- ***Clustering*** elements of a visual scene:
    - ***continuity***: grouping together edges/contours with the same orientation ("good continuation")
    - ***closure***: completing figures that have gaps
    - ***illusory contours***: perceiving contours and cues to depth, even when they don't exist
    - ***Common fate***: parts of the visual scene that move together are grouped together

Object constancy

- the brain correctly perceives objects as constant despite sensory data that could lead it to think otherwise.
- Mostly an object's size, shape, colour or lightness doesn't change even when different cues are there.
- To perceive any of these four constancies, we need to understand the relationship between the object and at least one other factor:
    - ***size constancy***, we need to know how far away the object is from us.
    - ***shape constancy***, we need to know what angle or angles we are seeing the object from.
    - ***colour constancy*** and ***lightness constancy***, we need to compare the wavelengths of light reflected from the object with those reflected from its background.

Face perception:

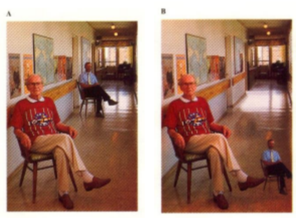
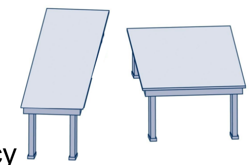
- We are well able to perceive and interpret facial expressions → any pattern with facelike qualities will look like a face.
- ***fusiform gyrus*** in the right hemisphere is important for perceiving faces.
- People with ***prosopagnosia*** cannot tell one face from another, but they are able to judge whether something is a face or not and whether that face is upside down or not.
    - → Suggests that we process faces differently than other objects.
- Psychologists debate whether faces are processed in a special way because they are important social signals or because they are objects with unique properties (***expertise hypothesis***).

Perception of depth

- Binocular depth cues → available from both eyes together and contribute in bottom-up processing. - only useful for nearby objects
    - Binocular disparity
        - caused by the distance between two eyes, each eye receives a slightly different retinal image, used to compute distances to nearby objects (stereoscopic vision).
    - Convergence
        - the way the eye muscles turn the eyes inwards to view nearby objects (the brain knows how much the eyes are converging)
- Monocular depth cues (pictorial depth cues),
    - available from each eye alone

    - Provide organisational information for top-down processing (perceive depth with one eye closed)
    - e.g.,
        - Occlusion
        - Relative size
        - Familiar size
        - Linear perspective
        - Texture gradients
        - Position relative to horizon
        - Shadow
        - Motion parallax

Perception of size
- Size perception depends on perceived distance

Perceptual constancies
- We correctly perceive different properties of distal stimuli despite continuously changing proximal stimuli.
    - Size constancy 
    - Shape constancy 
    - Lightness constancy
    - Colour constancy

18/09/2023

L5 (PS 6)

Learning

- ***Learning*** is a relatively enduring change in behaviour resulting from experience. (***adaptation***)
- Learning and memory are related
- ***Social learning*** = acquiring or changing a behaviour after verbal instruction or exposure to another individual performing that behaviour. (dode acteur op station)

Three types of learning:

1) Nonassociative learning
- repeated exposure to a single stimulus/event causes change.
- ***Habituation***
    - A *decrease* in an innate response to a frequently repeated stimulus, relies on memory

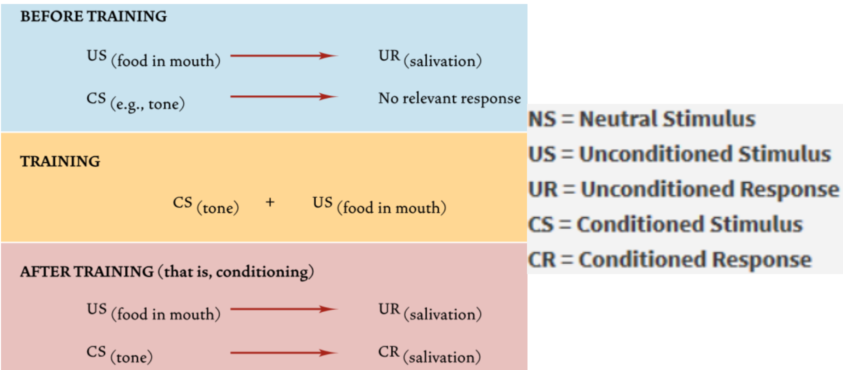
        - Allows a better ability to focus on what is new by ignoring that which has already become familiar
    - Opposite is ***dishabituation*** = an increase in a response because of a *momentary* change in something familiar
    - (getting used to bird song (habituation), being alarmed when they stop (dishabituation))
- ***Sensitisation***
    - An *increased* response to a stimulus after repeated exposure (zoals na een tijdje beter kunnen horen als iemand de trap opgaat ofzo)
    - Often threatening/pain stimuli (getting alarmed by burning smell).
        - It's a more generalised reactivity
    - Opposite is ***desensitisation*** = the extinction of sensitisation
- caused through alteration of neurotransmitter release in aplysia.
    - reduction of release causes habituation, increase causes sensitization.

<u>2) Associative learning</u>

- Understanding how stimuli/events are related, linking them through conditioning. two events always occur together → you learn they are linked

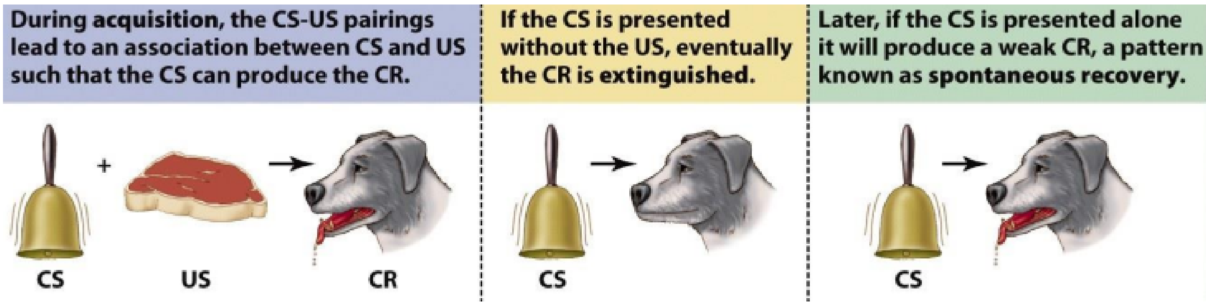
Classical conditioning

- A form of learning in which a neutral stimulus comes to elicit a response when it is associated with a stimulus that already produces that response (Pavlov met de hond)
- ***unconditioned response*** (UR): a response that does not have to be learned (saliva



production)
- ***unconditioned stimulus*** (US): the stimulus that elicits the response, without learning (food)
- ***conditioned response*** (CR): a response to a stimulus that had to be learned (formed an association) (eg: dog hearing the metronome that always clicked before the food came)
- ***conditioned stimulus*** (CS): the stimulus that elicits the response after learning (metronome)
    - Usually the conditioned response is weaker than the unconditioned one (less saliva).
- ***Second-order conditioning***
    - CS is not associated directly with US, but with another CS that is already associated with US (after metronome, a black square appeared and after a while dogs reacted with salivation just to the square).

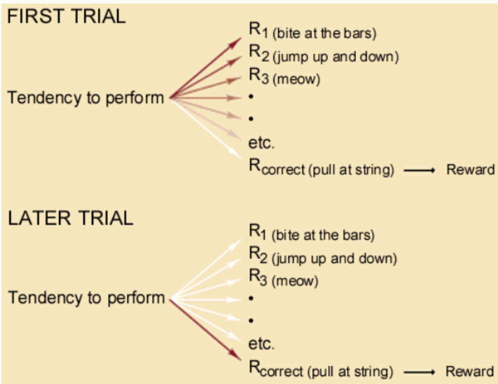
    - When a CS is paired with a new S, the new S produces CR
- ***Acquisition***
    - The formation of the association between the US and CS
- ***Extinction***
    - The association between the CS and CR can be eliminated by repeatedly presenting the CS alone
- ***Spontaneous recovery***
    - After extinction and a resting interval



- ***Rescorla-Wagner model***: learning is determined by the extent to which an US is unexpected or surprising.
    - An animal learns to expect that some predictors (potential CSs) are better than others.
    - ***Prediction errors***: something other than expected happens.
        - ***positive prediction error***: something better than expected happens / event happens without being expected, this strengthens the association (causes a great dopamine activity)
        - ***negative prediction error***: expected event does not happen, this weakens the association (reduction in dopamine activity)
- ***Stimulus generalisation***
    - occurs when stimuli similar but not identical to the CS produce the CR. (eg: different environments like light, temperature)
- ***Discrimination***
    - occurs when an animal learns to differentiate between two similar stimuli if one is consistently associated with the US and the other is not. (differentiate similar plants when one is poisonous)
    - CS+ → reinforced stimulus
    - CS- → unreinforced stimulus
- BUT an organism cannot learn *every* association
    - Biological constraints → ***biological preparedness*** = animals are genetically programmed to fear specific objects (fear of snakes)
- BUT → conditioning requires
    - ***Contiguity*** (together in time) - such that learning only occurs when the CS is presented *before* the US
    - ***Contingency*** - the CS should be informative with respect to the chance that the US is offered
- Thus, the CS functions as a sort of predictor of the US!
    - The CR is a reaction in preparation for the US.
    - This can also lead to compensatory reactions
        - Drugs
- Classical conditioning is a relatively passive and automatic process.

Operant conditioning (Instrumental conditioning)

- The *consequence* of an action/response determines the likelihood that it will be performed in the future.
- Behaviours represent a way to attain something (a reward) or avoid something (a punishment). They are done for a purpose.
- Learned is the relationship between response and the consequence (reinforcer/reward or punishment).
- ***Law of Effect*** (Thorndike, 1898)
    - any behaviour that leads to a "satisfying state of affairs" is likely to occur again; any behaviour that leads to an "annoying state of affairs" is less likely to occur again.
    - Behaviour is governed by its consequences.
    - Performance is strengthened if it's followed by a reward and weakened if it's not.
        - I.e., the tendency to give response



- ***Behaviourism*** = a psychological approach that emphasises environmental influences on observable behaviours
    - Certain incentives can change behaviour, behaviour occurs because it's been reinforced (B.F. Skinner)
- ***Shaping***
    - behaviour can be shaped by reinforcing behaviour that is similar to the desired one
    - Successive approximations
        - The reinforcer can't be provided until the animal displays an appropriate response, so reinforcing successive approximations eventually produces the desired behaviour.
- ***Reinforcers*** = a stimulus that occurs after a response and increases the likelihood that this response will be repeated
    - Primary (biological needs, food) and secondary reinforcers (not biological needs, money)
    - Positive and negative reinforcers (removal of unpleasant)
        - P: the administration of a stimulus to increase the probability that behaviour is being repeated, a reward.
        - N: the removal of an unpleasant stimulus to increase the probability of a behaviour being repeated.
    - ***Reinforcer potency***: some reinforcers are more powerful than others (food is

more desirable when you're hungry than when you are not).
        - ***Premack Principle***: a more-valued activity can be used to reinforce the performance of a less-valued activity
- ***Punishment*** = reduces the probability that a behaviour will recur
    - Positive and negative punishment (removal of pleasant)
        - P: the administration of a stimulus to decrease the probability of a behaviour recurring
        - N: the removal of a usually pleasant stimulus to decrease the probability of a behaviour recurring
    - Punishment is not always effective. learn how to avoid it, rather than how to behave properly.
    - can also evoke negative emotions, associated with the person giving the punishment. Positive punishment is very *ineffective*. Negative punishment can *change behaviour*, but reinforcing good behaviour is more effective.
- ***Continuous reinforcement*** = behaviour is reinforced each time it occurs. It can lead to fast learning, but the effect probably won't last.
- ***Partial reinforcement*** = intermittent reinforcement of behaviour, the effect depends on the type of ***schedule***:
    - ***Ratio schedule:*** is based on the number of times the behaviour occurs
        - Fixed-Ratio (FR), e.g., food pellet after 4 responses
        - Variable-Ratio (VR), reinforcement after unpredictable amount of responses (e.g., food pellet after on average 4 responses)
    - ***Interval schedule***: based on a unit of time
        - Fixed-Interval (FI), e.g., food pellet after 1st response after 4 minutes
        - Variable-Interval (VI), after unpredictable amount (e.g., food pellet after 1st response after on average 4 minutes)
- ***Partial-reinforcement extinction effect***
    - the greater persistence of behaviour under partial reinforcement than under continuous reinforcement.
    - The less frequent the reinforcement is during training, the greater the resistance to extinction.
- BUT an organism cannot learn any association
    - Biological constraints → built-in predispositions
    - "Never try to teach a pig to sing; it wastes your time and annoys the pig."
- BUT → learning also occurs in the absence of responses.
    - Learning involves not only a change in behaviour but also the acquisition of new knowledge (Tolman, 1948).
- BUT → conditioning requires
    - ***Contiguity*** (together in time)
        - learning only occurs when the reinforcer or punishment immediately follows the response.
    - ***Contingency***
        - The reinforcer or punishment should occur more likely after the response than otherwise.
        - Without contingency → learned helplessness
- ***Equipotentiality***: the principle that any conditioned stimulus paired with any unconditioned stimulus should result in learning

3) Observational/social learning

- The acquisition or modification of a behaviour after verbal instruction or exposure to another individual performing that behaviour (***Bandura's observational studies***)
    - two groups of children watching a movie with a doll ("Bobo") In the movie the first group watched, the doll was treated nicely. In the movie the second group watched, it was treated very aggressively. After that the children were allowed to play with that doll, the second group was twice as aggressive with the doll.

Modeling (demonstration and imitation)

- the imitation of observed behaviour
- influenced by the attractiveness, status and similarity to ourselves of the model.
- Depends on the physical capability to imitate the model
- (watching an athlete doesn't make us one).

Vicarious learning

- learning the consequences of an action by watching others being rewarded or punished for performing the action.

Instructed learning:

- acquiring associations and behaviour through verbal communication.

What do we know about the mechanisms of observational learning in the brain?

- Mirror neurons
- Social learning of cultural associations between Black race and criminality is proposed to contribute to racial bias in the U.S. legal system.

Dopamine

- An important component of the neural basis of reinforcement
- The dopamine release sets the value of the reinforcer (food tastes better when you're hungry)
- BUT there is a difference between wanting (desiring or craving a substance, maybe without even liking the consume itself) something and liking (the subjective sense of pleasure a user receives from consuming the substance) something.
- Through conditioning, secondary reinforcers can also activate dopamine receptors.
- The neurotransmitter dopamine provides one neurobiological basis for learning from prediction errors.
- Dopamine release increases after positive prediction error (unexpected reward) and decreases after negative prediction error (expected result did not happen).

Fear

- ***Phobia***: an acquired fear that is out of proportion to the real thread.
- ***Fear conditioning***: classical condition to fear neutral objects
- ***Counterconditioning***: overcoming a phobia by being presented with scary objects in combination with more pleasant things.
- People also learn fear by observation (watching others being scared) or by instruction (being told something will cause harm).

21/09/2023

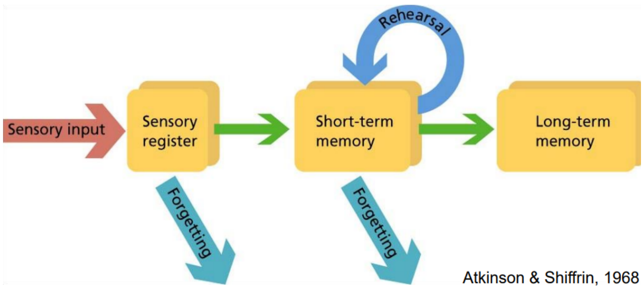
L6 (PS 7)
Memory

- Memory is the ability to store and retrieve information, to take information from experiences and retain it for later use.
    - Memory involves the creation or strengthening of neural circuits

The study of memory

- Perception
    - The organisation of current stimuli
- Memory
    - The organisation of stimuli from the past (ideas and events)
    - The nervous system's capacity to retain and retrieve skills and knowledge
    - We have multiple memory systems, and each memory system has its own "rules"
- Memory operates over time in three phases
    1. Encoding
    2. Storage and consolidation (in memory traces)
    3. Retrieval
        - Recall
        - Recognition

Three memory systems: ***stage theory of memory***



- Sensory memory, short-term or working memory, and long-term memory are differentiated by the length of time the information is retained in memory.

Sensory memory

- Sensory memory feeds our active working memory, recording momentary images, sounds, and strong scents.
- But sensory memory, like a lightning flash, is fleeting.
- ***Iconic memory***:
    - A momentary sensory memory of visual stimuli; a photographic or picture-image memory lasting no more than a few tenths of a second.
- ***Echoic memory***
    - A momentary sensory memory of auditory stimuli; if attention is elsewhere, sounds and words can still be recalled within 3 or 4 seconds.

Short-term memory/Working memory

- Represents what you are consciously focusing on any time.
- Can actively retain and manipulate multiple pieces of temporary information from different sources.

- Information remains there for 20-30 seconds → disappears unless rehearsed or thought about.
- Working memory can only hold a limited amount of information (***memory span***)

Long-term memory
- Longer duration and greater capacity than STM
- Information enters permanent storage through rehearsing or helpfulness to adapt to the environment (we want to store only useful information)

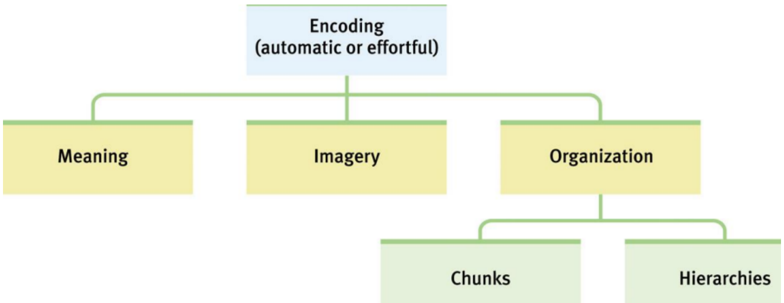
Serial position effect:
- The ability to recall items from a list depends on the order of presentation.
- Better memory of items at the beginning of the list: ***primacy effect***. (get rehearsed the most)
- Better memory of items at the end of the list: ***recency effect***. (still in working memory)
- Long-term memory is distinct from working memory. When we think about our memories, information from long-term memory enters working memory to be actively processed. And to chunk information in working memory, we need to form meaningful connections based on information stored in long-term memory

<u>1) Encoding</u>
- The transformation of a perceived stimulus or event into a memory.
- ***Dual-coding hypothesis***: information that can be coded verbally and visually will be remembered more easily than information that can be coded only verbally.
- Through rehearsal: STM → LTM
    - ***Maintenance rehearsal***: simply repeating the item over and over.
    - ***Elaborative rehearsal***: links new information with existing knowledge.
    - Elaborative rehearsal leads to deeper encoding and better recall.
- Memory is an active process requiring attention
- Modern memory theory
    - STM → working memory
        - Information one is thinking about
    - Long-term memory (LTM)
        - Information, stored during longer intervals, which is currently not actively used.
- Which encoding processes promote later recall?
    - Depth of processing
        - Shallow encoding processing (surface form)
        - Deep encoding processing (meaning)
    - Understanding
    - A ***schema*** may help you to understand and subsequently remember information contents.
        - Cognitive structures in semantic memory that helps with perceiving, organising, understanding, and using information.
- Good encoding provides an effective means for retrieval later.

Encoding: getting information in
- ***Effortful processing***

        - Requires attention and conscious effort.
            - Your new pin pass code
- ***Automatic processing***
    - Unconscious encoding of incidental information
        - Space, time, and frequency
    - And of familiar or well-learned information
        - Sounds, smells, and word meanings
        - Route to school

Encoding hierarchy

- 

Encoding by meaning

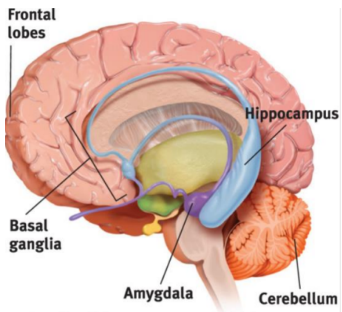
- Encoding meaning (semantic encoding) results in better recognition later than visual or acoustic encoding

Visual encoding

- Mental pictures (imagery) are a powerful aid to effortful processing, especially when combined with semantic encoding.

<u>2) Storage</u>

- After encoding → storage in memory traces
    - Memory consolidation
        - The neutral process by which encoded information becomes stored in memory → transient and fragile potential memory contents are transformed into a more permanent state
        - ***Long-term potentiation (LTP):*** Strengthening of a synaptic connection during consolidation, making the postsynaptic neurons more easily activated by presynaptic neurons
            - ***NMDA receptor (glutamate)*** = LTP leads to more NMDA receptors in postsynaptic neuron → increased responsivity to released glutamate by presynaptic neuron
        - ***Gradual consolidation*** allows stored memory to be modified
        - After rehearsal → ***reconsolidation*** (might corrupt)
            - Original memory can be modified or strengthened
            - Triggered when something relevant and new about older memory is presented



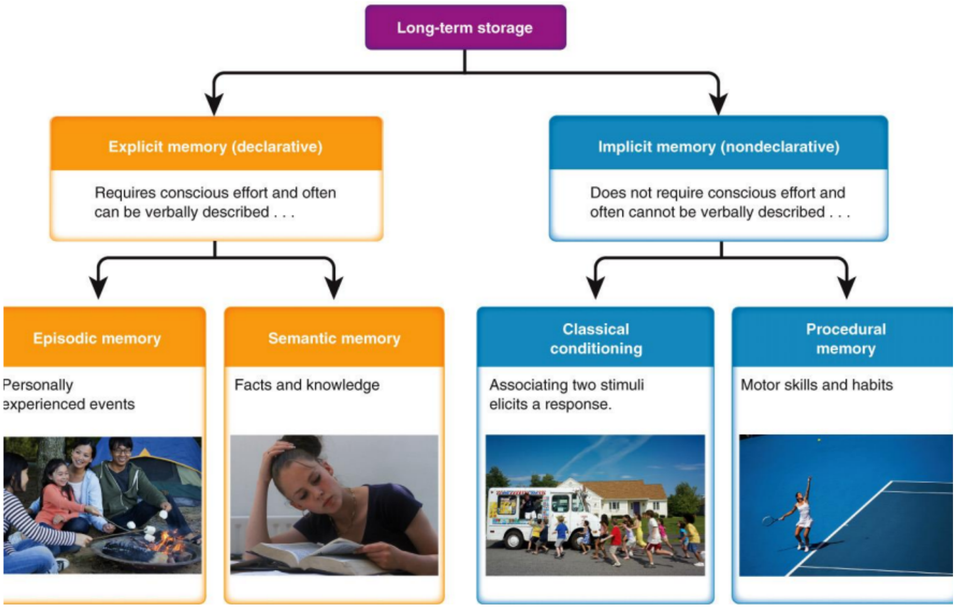
Where is the storage?

- Medial temporal lobes and hippocampus: explicit memory formation

- Cerebellum and basal ganglia: implicit memory formation
- Amygdala: emotion-related memory formation

Damage to hippocampus
- Disrupts the formation and recall of explicit memories.
- ***Memory consolidation***
    - The neural storage of a long-term memory.
    - Sleep supports this.

Stress Hormones & Memory
- Heightened emotions (stress-related or otherwise) make for stronger memories
- ***Flashbulb memories*** are vivid episodic memories of what we were doing during emotionally significant moments or events.
    - Not perfectly accurate, but people remember central details of the event and what they were doing
- Stress hormones make more glucose energy available to fuel brain activity



- ***Explicit (declarative) memory***
    - Retention of facts and experiences that we can consciously know and “declare”.
    - ***Episodic memory***: a person’s past experiences, includes time and space of the experience.
    - ***Semantic memory:*** knowledge of facts, independent of personal experience (you might not remember when or where you learned it, but you know it)
- ***Implicit (non-) memory***
    - Memory that is expressed through responses, actions, or reactions. (Priming, procedural, classical conditioning, non-associative learning)
    - Nondeclarative memory
        - ***Procedural memories***: skilled and goal-oriented behaviours that become automatic, such as motor skills, cognitive skills, and habitual behaviours.
        - ***Priming:*** a facilitation in a response to a stimulus due to recent experience with that stimulus or a related stimulus.

            - ***Perceptual priming:*** a response to the same stimulus
            - ***Conceptual priming:*** a response to a conceptually related stimulus

<u>3) Retrieval</u>

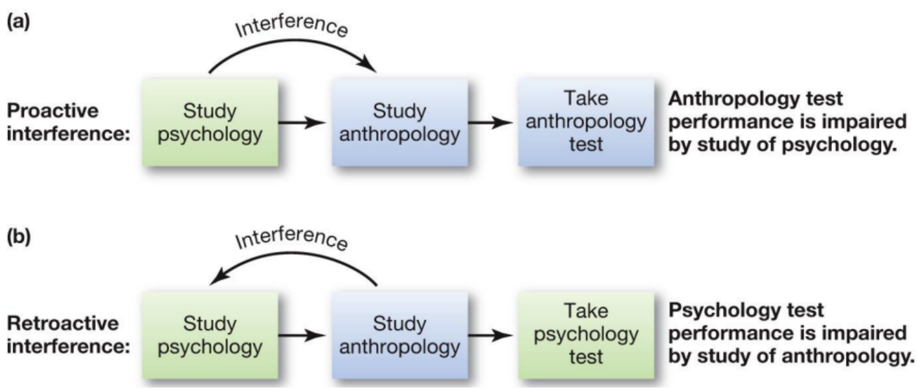
- Recognition
    - The person must identify an item amongst other choices.
        - A multiple-choice test requires recognition
- Recall
    - The person must retrieve information using effort
        - A fill-in-the-blank test requires recall.
- ***Retrieval practice***
    - A strategy of bringing information to mind by deliberately trying to recall it.
    - results in the type of retrieval that is more likely to trigger reconsolidation, strengthens the memory more
    - A ***retrieval cue*** can be anything that helps a person recall a memory
- Good retrieval cues
    - Re-create the context in which the original learning occurred → context reinstatement
        - E.g., experiment divers
    - Re-create the state of mind in which the original learning occurred
    - A richer encoding context results in better recall. → ***encoding specificity principle***
        - Any stimulus encoded along with an experience can serve as a retrieval cue during recall.
    - ***Context-dependent memory***: occurs when aspects of the encoding context match the retrieval context.
    - ***State-dependent memory:*** occurs when a person's internal state matches during encoding and recall.
        - also applies to internal states brought on by drugs or alcohol.

Forgetting

- An inability to retrieve information due to poor encoding, storage, or retrieval.
- Human memory is biassed, distorted, and prone to a great deal of forgetting. Memory is a constantly active process, involving consolidation and reconsolidation of information manipulated in working memory.
- Memories are forgotten over time. The rate of forgetting is rapid at first and slows over time. If a memory lasts 5 years, it is likely to last a lifetime
- Re-learning something takes less time than it did to learn at first, the difference between the original learning and relearning is called ***savings***.
- ***Blocking***
    - The temporary inability to remember something
    - Like in the ***tip-of-the-tongue phenomenon***. Blocking often occurs because of interference from words that are similar in a way
- Not being able to forget is as maladaptive as not being able to remember. Normal forgetting helps us retain and use the most meaningful and important information.
    - ***Persistence:*** the continued recurrence of unwanted memories (PTSD).

When memory fails

- ***Amnesia***: a deficit in long-term memory -resulting from disease, brain injury, or psychological trauma- in which the individual loses the ability to retrieve vast quantities of information.
  - ***Retrograde amnesia***: a condition in which people lose past memories
  - ***Anterograde amnesia***: a condition in which people lose the ability to form new memories
- An inability to remember
  - Encoding failure
    - We cannot remember what we do not encode. Information never enters long-term memory.
  - E.g., absentmindedness, memory decay
- Memory reconstructions (distortions of memory)
  - E.g., bias, misattribution, suggestibility
- ***Absentmindedness:***
  - The inattentive or shallow encoding of events
    - During most of our daily activities we are consciously aware of only a small portion of both our thoughts and our behaviours.
- ***Memory decay***:
  - Forgetting over time
  - Why?
    - Decay of memory traces with time (caused by metabolic processes)
    - Interference
      - After sleep less forgetting than after wakefulness
      - Proactive and retroactive interference



- ***Memory bias***:
  - The changing of memories over time so that they become consistent with current beliefs, knowledge or attitudes
  - People often revise memories when they change attitudes and beliefs.
    - Flashbulb memories can be biassed
  - Tends to cast memories in a favourable light.
- ***Source misattribution***:
  - Memory distortion that occurs when people misremember the time, place, person, or circumstances involved with a memory.
  - E.g, ***false fame effect***
    - tendency to think names they have heard before are names of famous people

    - E.g, ***sleeper effect***
        - Unpersuasive argument from a questionable source becomes credible over time as the source is forgotten.
    - E.g, ***Source amnesia***
        - occurs when people have a memory for an event but cannot remember where they encountered the information
    - E.g, ***cryptomnesia***
        - occurs when people think they have come up with a new idea yet have retrieved a stored idea and failed to attribute the idea to its proper source.
- ***Suggestibility***:
    - The development of biassed memories based on misleading information
        - Different wordings of questions after an event may alter the participants' memories of the event → Loftus
        - Memories can be distorted, or even implanted, with false information
        - Memories can be altered by the intrusion of generic knowledge or semantic associations.
        - could play a role in eyewitness testimony, as research shows that eyewitnesses can develop biassed memories based on leading questions

Memory construction
- While tapping our memories, we filter or fill in missing pieces of information to make our recall more coherent.
- ***Misinformation Effect***: Incorporating misleading information into one's memory of an event.
    - After exposure to subtle misinformation, many people misremember.
    - Guessed details are absorbed into our memories

Children's Eyewitness Recall
- Children's eyewitness recall can be unreliable if leading questions are posed.
- However, if cognitive interviews are neutrally worded, the accuracy of their recall increases.

False memories
- ***False memories*** are created from the natural tendency to form mental representations of stories → imagining an event happening, you form a mental image of the event, and might later confuse that mental image with a real memory.

25/09/2023

L7 (PS 8)
Thinking

What is thinking?
- Cognitive psychology is the study of mental functions such as intelligence, thinking, language, memory and decision-making
    - ***Cognition*** = mental processes that are involved with the accumulation of knowledge and meaning through thoughts and experiences

1) Mental representations

Cognitive psychology

- Originally based on two ideas of thinking
    - Knowledge of the world is saved in the brain in representations
    - Thinking is the mental manipulation of these representations
        - We use representations to understand objects we come across in the environment
- Components of knowledge:
    - Mental representations
        - Analogical representations (foto van kat)
            - Mental representations that have physical traits of objects
            - Analogical representations are dissimilar from pictures
        - Symbolic representations (woord kat)
            - Abstract representations that do not match with the physical traits of objects or ideas
    - Together they form the basis of human thinking, intelligence and the ability to solve complex problems of daily life.

Categories

- Memory systems → organised in such a way that we can quickly retrieve information when we need it.
    - E.g., when asked what a violin is → musical instrument (category)
- ***Categorisation*** = The grouping of things on the basis of shared elements
    - This mental activity lessens the amount of knowledge we keep in the memory and thus is a efficient way of thinking
    - We do have to save unique knowledge for every component in a group

Concepts

- ***Concept*** = a category/class of related items
    - By allowing us to organise mental representations around a common theme, a concept saves us from having to store each instance of an object separately
- How do we represent concepts in our mind such that we can call up information quickly? Two leading models:
    - ***Prototype model***: a way of thinking about concepts: within each category there is a best example -a ***prototype***- for that category.
    - ***Exemplar model***: argues that every concept does not have one best representation. Instead, all the examples that you have actually encountered, form the concept.
        - This model assumes that people form a vague representation of a concept through experience because there is no single representation of a concept.
        - Only one way in which people's thoughts are unique
    - Explains how we classify objects we encounter and how we represent those objects in our minds.

Conceptual thinking in the brain

- Different kinds of knowledge is differently coded
- Patterns of brain activity can be used to identify aspects of conscious experience

    - E.g., whether people look at faces or objects, feel angry or happy, etc.
- Imagery studies have shown that different categories of objects are represented in different brain regions based on our perception of the objects.
    - Thinking of animals: activates visual areas → just how they look allows us to organise them as ‘animals’

Schema’s
- ***Schema’s*** = helping to perceive, organise, understand and process information
    - We can use it because common situations have consistent rules and because people have specific roles within situational contexts
- ***Scripts*** = common type of schema, which helps us understand the sequence of events in certain situations
    - It controls behaviour over time within a situation
- Reason for schemas and scripts:
    - Organise information and common scenarios.
    - Helps us understand and process information more efficiently
- However, sometimes schemas have unintended consequences
    - E.g., strengthening stereotypes

<u>2) Decision making</u>
- ***Decision-making***:
    - attempt to choose the best alternative from several options. Typically, we identify important criteria and determine how well each alternative meets these criteria
- ***Judgement***
    - Deciding on the likelihood of various events using incomplete information

Traditional decision research
- ***Rational choice***:
    - A single “correct” choice after a cognitive evaluation
- Behavioural violations of “rational choice models”

Decision making
- ***Normative decision theories*** define how people should make decisions.
    - ***Expected utility theory***
        - Decisions should be made to maximise the outcome.
        - This should depend on
            - The values of the alternative options
            - The probabilities associated with the alternative options
            - E.g., a lottery ticket costs $10 and p(winning $1mil) = 10^-6
                - EV (buying) = ((1 - 10^-6) * - $10) + (10^-6 * $1mil) = -$9
                - EV (not buying) = (1 * $0) + (0 * $1mil) = %0
- ***Descriptive decision theories*** indicate how people actually make choices
- ***Heuristics***
    - Shortcuts (rules of thumb or informal guidelines) used to reduce the amount of thinking that is needed to make decisions
        - Heuristic thinking occurs often unconsciously

        - Heuristic thinking allows us to decide quickly and concentrate on other things
        - Heuristics can also result in errors or faulty decisions
    - ***Conformation bias***: only focus on information that confirms their idea
    - ***Hindsight bias***: making the explanation after the fact.
    - Other common heuristics that distort decision making:
        1. Relative comparisons (Anchoring and framing)
        2. Availability
        3. Representativeness.

Relative comparisons

- ***Anchoring***
    - The tendency, in making judgments, to rely on the first piece of information encountered or information that comes most quickly to mind
- ***Framing***
    - In decision making, the tendency to emphasise the potential losses or potential gains from at least one alternative
    - Decision is influenced by the way the voice is designed.
- ***Framing effect***
    - Asian disease problem (Tversky & Kahneman, 1981)
        - 600 people in danger

	Positive Frame	Negative Frame
Treatment A	72% Saves 200 lives	400 people will die
Treatment B	33% chance of saving all 600 people 66% possibility of saving no one	%33 chance that no people will die 66% probability that all 600 people will die

        - 78%

Gains & losses

- ***Risk aversion*** = preferring a sure gain to a risky gain
- ***Loss aversion*** = more sensitivity to potential losses than potential gains

Heuristics

- ***Availability heuristic***
    - Things that pop up easily in our mind tend to guide our thinking
    - Vivid, recent or memorable events/memories come easily → can bias our thinking
    - often uncommon events but more emotional ones or sensational, so come to mind more readily
- ***Representativeness heuristic***
    - Deciding based on how typical an individual, object, or event belongs in a certain category
    - We use it when we base a decision on the extent to which each option reflects what we already believe about a situation
- ***Affective forecasting heuristics***
    - Decision-making by what benefits you emotionally the most in the future.

- ***Affective predictions***: The tendency for people to overestimate how events will make them feel in the future
- ***The paradox of choice***
  - When too many options are available, especially when all of them are attractive, people experience conflict and indecision
- A good strategy is to *satisfice* rather than *optimise* your decisions

Emotions in decision making
- Emotions can also influence decision making
- After a negative event, people engage in strategies that help them feel better. They rationalise why an event occurred and minimise the importance of the event.
  - Adaptive; Protect mental health → understanding an event helps reduce its emotional impact
- When rating their overall satisfaction with life, people in a good mood were more likely to rate their lives as satisfactory, while people in a bad mood gave lower overall ratings

Affective states
- Unlike emotions that are integral to choices, affective states completely unrelated to the choice can also distort decisions.
  - E.g., snapping at something because you are angry about something else.
- ***Endowment effect***
  - Tendency to value things we have more than we would pay to buy them

<u>3) Problem solving</u>
- Using information to achieve a goal → ***Goal-oriented***
- Problems come in two forms:
  1. Easy to define problems (e.g., where are my keys)
  2. Undefined problems (e.g., what kind of job should i aim for)
- To fix the problem:
  - Make a plan to get one achievable goal
  - Devise strategies to overcome obstacles to implement the plan
  - Evaluate the results to see if the goal has been achieved
- Obstacles to problem solving
  - Complexity
  - ***Mental sets***: Problem-solving strategies that have worked in the past.
    - Difficult to unlearn.
    - Comes from having fixed ideas about the typical functions of objects. =***Functional fixedness***.
    - Instruction / practice
    - Perceptual organisation of the problem
- Overcoming obstacles to solutions
  - Understanding → formulate subgoals
    - Steps to solve a specific goal
  - ***Restructuring***
    - A new way of thinking about a problem that contributes to its solution.
    - The new view can offer a solution that was not visible under the old problem structure.

        - Use an algorithm
            - ***Algorithm***: Guideline, which if followed correctly will always yield the correct answer.
        - Use an analogy → ***analog problem solving***
            - Using a strategy that works in one context to solve a problem that is structurally similar.
            - Researchers have found that participants who solve 2 or more analogous problems develop a schema that helps them solve similar problems.
        - Wait for a sudden insight.
            - Wolfgang Köhler
                - Chimpanzees get banana outside of the cage: 2 sticks, both not long enough
                - Insight to tie the sticks together. Transferred this solution to similar problems and quickly solved them
            - Norman Maier
                - 2 strings hanging and a table with stuff on it in the corner. Impossible to grab both strings at the same time.
                - Solution was to tie the pliers to one string of rope and use it as a sling.

<u>4) Language</u>

- ***Language*** = a system of communication using sounds and symbols according to grammatical rules.
    - ***Syntax*** (rules): The system of rules that govern how words are combined to form sentences and how sentences are combined to form sentences.
    - ***Semantics*** (meaning): The study of the system of meanings underlying words, phrases, and sentences.
    - → Hierarchical structure
        - Sentences can be split into smaller sentences. And sentences are broken down into words. Words are divided into sounds.
        - ***Morphemes*** = words
            - The smallest units that have meaning, including suffixes and prefixes
        - ***Phonemes*** = letters/sounds
            - Basic sounds of speech, the building blocks of language
            - Phonemes signal meaningful differences between words.
        - “Kissed” consists of 2 morphemes → "Kiss" and "ed".
        - This word consists of 4 phonemes → “k”, “i”, “s”, “t”.

Social and cultural influences

- Environment has a major influence on language acquisition.
- ***Creole:*** a language that evolves over time through the mixing of existing languages.
    - Develops from rudimentary communication, as populations that do not speak different languages try to understand each other.
    - ***Pidgin:*** mixed words from each other's language.

Development of language

- Development occurs in an orderly way
- 2 factors are needed to support normal language acquisition:
    - Children need someone to communicate with.
        - Social communication is a critical and determining factor in language development.
    - Children need to be exposed to language during critical periods of their development. (In the womb, just born, etc.)
- Development occurs in an orderly way:
    - First verbal sounds: Crying, growling, breathing, gurgling.
    - Laugh.
    - Babbling, using consonants and vowels.
    - Babbling syllables.
    - First words (often items from their environment and simple words).- Merge words. Finally sentences.
- ***Telegraphic speech*** (Roger Brown):
    - Tendency of toddlers to speak using rudimentary sentences that lack words and grammatical markers.
    - But follow logical syntax and convey significant meaning.
- Newborns can discriminate all the phonemes but lose this ability after 6 months
- Children understand that speakers usually think about what they are looking at.
    - If a parent looks at the toy when saying a new name, the child will assign the name to that object.
- ***Phonics (sounds)***: A method for teaching people to read that focuses on the association between letters and their phonemes.
    - People with ***dyslexia*** have difficulty reading, spelling and writing, even though they have a normal intelligence level.
        - Problem: Inability to hear, identify and use phonemes.
            - Dyslexia may have a genetic component.
            - The environment also plays a role.

Overgeneralizations
- children begin to use language in more ways → may overapply new grammar rules they learn.
- Children are able to use language effectively by observing patterns in spoken grammar and then applying rules to new sentences they have never heard before.

Acquisition of correct language
- System of ***operant reinforcement*** (Behaviourist Skinner):
    - Children are encouraged to correctly repeat what their parents say.
    - Speech that is not reinforced by parents is extinguished.
- BUT: Language acquisition doesn't work that way.
    - Parents do not correct their children's grammatical errors or continually repeat words and phrases to their children.
        - correct if the content of what they say is incorrect, but not if the grammar is incorrect.
- ***Universal grammar*** (Linguist Naom Chomsky):
    - *"All languages are based on man's innate knowledge of a set of universal and specific linguistic elements and relationships."* (Chomsky)

        - ***Surface structure***: In language, the sound and order of words.
        - ***Depth structure***: In language, the implied meanings of sentences.
    - People automatically/unconsciously transform surface structures into deep structures.
    - Chomsky: People are born with one ***language interweaving device***, which contains universal grammar.

Making sounds

- Every language is derived from a very limited set of phenomena.
- The human vocal tract has the capacity to make many more sounds than any language uses.
    - Languages differ from one another not only by the words that are used but also by the number of *phonemes* and the patterns of *morphemes*
- How we make sound:
    1. People speak by forcing air through the vocal cords.
        - ***Vocal cords***: Folds of mucous membranes that are part of the larynx, an organ in the neck, often called the voice box.
    2. The air passes from the vocal cords to the oral cavity.
        - There, jaw, lip, and tongue movements change the shape of the mouth and airflow, and change the sounds produced by the vocal cords.
- Sign language intertwines on an identical timetable for child development as spoken language.

Understanding

- Different brain areas work together to separate the relevant sounds into segments, which allow interpretation.
- Meaning plays an important role in this perception. When you listen to a language you are not fluent in, it can be difficult to separate it into segments.

Language in the brain

- ***Aphasia:*** Language disorder that results in deficits in language comprehension and production. (Often due to strokes in the left hemisphere of the brain)
    - Damage in Broca's area → ***Expressive aphasia*** (Broca's aphasia):
        - Interruption of ability to speak
        - generally understand what is said to them, can move their lips and tongue, but they cannot form words or join one word to another to form a sentence.
    - Damage in Wernicke's area → ***Receptive aphasia*** (Wernicke's aphasia):
        - Difficulties to understand the meaning of words
        - They can talk, but it doesn't follow grammatical rules and isn't logical.
        - Due to damage in an area of the left hemisphere where the temporal and parietal lobes meet
- a network of brain regions work together to facilitate language
    - For about 90% of people, Left hemisphere → most important for language.
        - ***global aphasia:*** no longer be able to produce or understand language, because of major damage to the left hemisphere
    - Right hemisphere → contributes to language,

        - processing the rhythm of speech
        - interpreting what is said, especially metaphors.

Language and thought
- ***Linguistic relativity***:The idea that language determines thinking. We can only think through language,
- Benjamin Whorf: language reflects how people think
    - culture determines language → determines how people form concepts and categorise objects and experiences.
- Another theory is that language influences rather than determines thinking.

25/09/2023

L8 (PS 8)

Intelligence

What is intelligence?

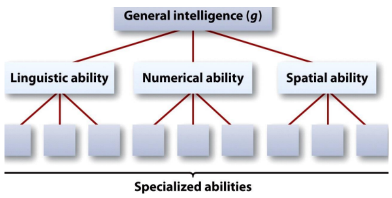
"The ability to reason, plan, solve problems, think abstractly, comprehend complex ideas, learn quickly, and learn from experience." (Gottfredson, 1997)
- One general ability?
- Different subtypes of abilities?

<u>1) Intelligence testing</u>
- The ***psychometric approach*** of measuring intelligence focuses on how people perform on standardised tests that assess mental abilities.
    - Examine what people know and how they solve problems
    - Approach influenced how we view intelligence in everyday life
- There are 2 main types of standardised tests:
    1. ***Performance tests***: Assess people's current level of skills and knowledge.
    2. ***Aptitude tests***: Trying to predict which tasks, and perhaps even which jobs, people will be good at in the future.

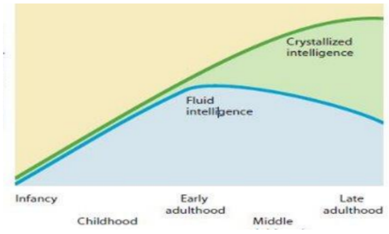
How to measure intelligence?
- Standardised psychometric tests
    - IQ = MentalAge / ChronologicalAge * 100 (Wilhelm Stem)
        - ***Mental age***: An assessment of a child's intellectual status compared to that of peers; determined by comparing the child's test score to the average score for children of each age.
        - IQ is measured differently in adults. Measured in comparison to the average adult and not to adults of different ages.
    - Deviation IQ
        - A person's IQ is considered in terms of deviation from the average.
- Tests for children
    - E.g., Binet-Simon Test or Stanford-Binet Test, Wechsler Intelligence Scale for Children (WISC)
    - ***Binet-Simon Test***: To identify children in the French school system who needed extra attention and special instruction.
        - Intelligence best understood as a collection of high-level mental processes.

        - With assistant (simon) developed a test to measure a child's vocabulary, memory, number and other mental skills. → ***Binet-Simon intelligence scale***
    - ***Stanford-Binet test***:
        - Lewis Terman (of Stanford University) modified the Binet-Simon test and established normative scores (average scores for each age).
        - remains the most widely used test for children in the US.
- Tests for adults
    - E.g., Wechsler Adult Intelligence Scale (WAIS / WAIS-R)
        - consists of 2 parts. Each part consists of different tasks that yield separate scores.
            - Verbal part measures aspects such as comprehension, vocabulary, general knowledge and test of working memory.
            - Performance part includes non-verbal tasks, such as arranging pictures in the correct order.
    - E.g., Scholastic Aptitude Test (SAT)
        - Group test
    - E.g., Raven's Progressive Matrices Test
        - Non-verbal test
- Intelligence tests
    - High reliability → Are results stable over time?
    - High predictive validity (but depends on the complexity of someone's job!) → Are they measuring what they really need to measure?

<u>2) The psychometric approach</u>

- An attempt to understand the nature of intelligence by studying the pattern of results obtained on intelligence tests.
    - One general ability or multiple specific abilities?
        - E.g., WAIS-R
        - Spearman (1904): general intelligence (g) + specific abilities
            - ***Factor analysis***: Statistical technique in which items that are similar are clustered (Factors).
            - With reference to this analysis Charles Spearman discovered that most intelligence items tended to cluster as one factor.
            - 

        - Cattell (1971): fluid vs. crystallised intelligence
            - General intelligence consists of two types:
                - ***Fluid intelligence***: understand abstract relationships, think logically without prior knowledge, reasoning, drawing analogies, and thinking quickly and flexibly.
                    - (Similar to working memory)
                - ***Crystallised intelligence***: knowledge acquired through experience, use it, and solve problems.

        - (Similar to long term memory)
        - Grows throughout the adult years.
    - Because both types are components of general intelligence, people who score high on one factor often also score high on the other.
    - Because different brain areas support fluid or crystallised intelligence, damage/abnormalities in a particular brain area may affect only some aspects of intelligence.
    - 

General intelligence
- ***General intelligence***: The idea that one general factor underlies intelligence.
- general intelligence influences important life outcomes (e.g., predicting performance in school and work) → Low general intelligence linked to earlier death, (heart disease, diabetes, stroke, Alzheimer's disease, …)
    - may be the result of different environmental forces.
        - People who do not perform well in academics → dangerous jobs.
        - People in less dangerous/higher paying jobs → better health care.
- Speed of mental processing
    - researchers found that the brains of highly intelligent people work faster than those of less intelligent people.
    - The relationship between general intelligence and mental speed appears to be associated with a longer lifespan in people with a high IQ.
        - Lower response time → Related to more health risks.

<u>3) Intelligence beyond the IQ test</u>
- Howard Gardner: each person has a unique pattern of intelligence, and no one should be considered smarter than others – just differently talented.
- Standard intelligence tests typically fail to capture “other types of intelligence”
    - Sternberg (1999): 3 types of intelligence:
        - ***Analytical intelligence***: solve problems and complete analogies,
        - ***Creative intelligence***: ability to gain insight and solve new problems – think in new and interesting ways.
        - ***Practical intelligence***: everyday tasks. E.g, good judge of people,
    - Salovey & Mayer (1990): emotional intelligence:
        - Linguistic, logical-mathematical, spatial, musical, bodily-kinesthetic, inter- and intrapersonal, and naturalistic intelligence

<u>4) The relation between IQ and cognitive performance</u>
- Different measures correlate with g
    - Simple RT and choice RT task
    - Discrimination task performance

        - Working memory capacity
        - Executive control
- Francis Galton:
    - that intelligence was related to the speed of neural responses and the sensitivity of the sensory/perceptual systems.
    - Smartest people had the quickest reactions and the sharpest observations.
    - He also speculated that they would have larger, more efficient brains.

<u>5) The roots of intelligence</u>

Nature (genetics) and nurture (environment) are intertwined in the development of intelligence.

- ***Epigenetics***: Environmental influences can alter gene expression to promote synaptic connections and processing efficiency in the brain.
- Genetic influences
    - There are thousands of genes that contribute to intelligence, and each individually has only a small effect.
    - Twin studies (identical twins versus fraternal twins)
        - Studies show that twins raised apart are very similar in intelligence
    - Adoption studies
- Environmental influences
    - Impoverished versus enriched environments
    - ***prenatal*** & ***postnatal factors***
        - ***Pre***: Factors during pregnancy.
        - ***Post***: After birth.
            - E.g. family, education, diet, cultural views, etc.
    - ***social economical status***
        - Growing up in a wealthier family → higher IQ and cognitive processes.
        - Because of intellectual opportunities.
    - ***Flynn effect***
        - genes have not changed much during that period, the increase must be due to environmental factors. (e.g., nutrition, healthcare, etc.)
- ***Heritability (H)***
    - A number between 0 and 1
    - 0 = everything due to environment, 1 = genetics

$$H = \frac{Genetic\ Variance}{Phenotypic\ Variance}$$

- Group differences
    - Men versus women
    - Black vs white

Nature or Nurture: Estimate of Heritability

- ***Heritability***
    - A *statistical estimate* of the proportion of *variance* in a given population that is due to genetics
    - Heritability of .6 = genetic influence explains about 60 percent of the observed variation among people
    - Some variables such as eye colour and height are highly heritable, other variables such as eating habits are less heritable.
- Heritability is not fate. It’s not inevitable
    - If you change the environment, you can change the traits of heritability.

    - BMI is heritable, but access to quality food and exercise have an impact.
- High heritability does not mean group differences are genetic.
    - The capacity for having a large vocabulary is considerably heritable, but every word in a person's vocabulary is learned in an environment.

02/10/2023

L9 (RM 1,2)
Scientific reasoning

Biasses

- ***Cognitive bias*** (intuition)
    - the human mind thinks imperfectly:
        - "makes sense":
            - We sometimes believe a story just because it is "common sense" or a "good story" when evidence goes against it
        - Info comes to mind easily
            - Conserving mental effort – "Mental shortcuts"
            - Availability heuristics
                - can lead us to overestimate any frequency (e.g., Pa complains you only talk to him for money – actually you talk a lot but asking for money is annoying so sticks out in mind)
            - We notice what is present more than we notice what is absent
- ***Motivational bias*** (intuition)
    - Sometimes we do not want to challenge our preconceived ideas:
        - We are motivated to think what we want to think
        - Seeing and accepting only the evidence that supports our belief
- ***Conformation Bias***
    - Cherry-picking evidence
- ***Bias blind spot***:
    - The belief that we are unlikely to fall prey to the cognitive biases previously described
- ***Hindsight Bias***
    - The "i-knew-it-all-along" phenomenon.
    - After learning the outcome of an event, many people believe they could have predicted that very outcome.
    - Everything seems common-place once explained.
    - Perceive order in random things
- ***Overconfidence***
    - Sometimes we think we know more than we actually know
- Both hindsight bias and overconfidence lead us to overestimate our intuition

<u>1) Fixation of belief</u>

- ***Belief:*** An attitude of some person toward a particular claim
    - I believe that "like attracts like as stated in the law of attraction".
- ***Claim*** (or ***statement***): True / False
- There is no direct connection between belief and truth
    - Believing that a given claim is true does not provide information about the truth or falsity of the claim itself.

- What are the methods to fix beliefs?
    - Non-scientific Methods:
        - Personal experience
            - Too much happens in everyday life; even if a change has occurred we cannot be sure what caused it
            - ***Confounds*** = potential alternative explanations
                - the things that might have caused a change and you didn't control for
                    - Hard to isolate variables in the real world
        - Intuition
        - Authority
    - Scientific Methods:
        - On the basis of ***Empirical research***: It reaches conclusions by combining:
            - Proper reasoning (i.e., use of logic arguments)
            - Systematic empiricism (i.e., use of verifiable evidence)

Solvable problems

- Science only deals with solvable problems
- Researchers can investigate only those questions that are answerable, given current knowledge and research techniques
- In sum, to be considered scientific:
    1. The questions addressed must be potentially solvable
    2. Observations must be systemic and empirical
    3. Research must be conducted in a manner that is publicly verifiable
- ***Pseudoscience*** involves evidence that masquerades as science but that fails to meet one or more of the three criteria used to define science.
    - Disseminated to general public via sources that are not peer reviewed
        - No reviews for precision or accuracy

<u>2) What is science?</u>

- Scientific method
    - A method to provide good justification for the belief that a claim or statement (hypothesis) is true or false.
- Goal of science
    - To find good justification for the belief that a claim or statement (hypothesis) is true or false
        - By means of a ***conditional argument***

Affirming the antecedent
If A then B
A
Thus B (valid)

Modus ponens
Premises: If it rains, I stay at home / It rains
Conclusion: I stay at home

Denying the consequent
If A then B
not B
Thus, not A (valid)

Modus tollens
Premises: If it rains I stay at home / I do *not* stay at home
Conclusion: It is not raining

Deductive versus inductive

- ***Deduction***

    - Specifying a conclusion about something specific on the basis of a general rule (assumptions/premises)
    - ***Deductive reasoning***:
        - Top-down, using things you already know (rationalism)
        - Infinity exists, given the fact you can always add a number to a previous one
- ***Induction***
    - Specifying a general rule based on specific observations
    - The conclusion is often plausible but not inescapable
    - ***Inductive reasoning***:
        - Bottom-up, using observations (***empiricism***)
        - Infinity cannot (never) be proven because nobody can count to infinity

Deductive reasoning - Aristoteles
- ***Syllogism***:
    - 2 premises → conclusion
        - Does the conclusion hold, given the premises?
        - Is not the same as true or false!
    - Two types of syllogisms:
        - Categorical syllogism (All A are B)
        - ***Conditional syllogism*** (If A then B)
            - If P then Q
Not P
Therefore not Q

Laws & Theories
- ***Laws:*** Describe
    - Newton's Law of Universal Gravitation describes *WHAT* an object will do when you drop it
- ***Theories:*** Explain
    - Einstein's Theory of General Relativity explains *WHY* it falls
    - Theories dictate important variables
    - ***Implicit Theory***
        - Everyone has one → based on our individual experiences
        - Shapes how we view the world; very difficult to change
    - Theories are Models
        - ***Models:*** Formalise to explain or predict
        - Variables from the theory are used to create the model

The four scientific cycles
- Cycle of 4 cycles to approach science

1. ***theory-data cycle (Empirical cycle)***
Theory → Research Questions → Research Design → Hypotheses → Data → Back
    - ***Theory*** = A set of statements that describes general principles about how variables relate to one another
    - ***Hypotheses*** (prediction) = a way of stating the specific outcome the researcher expects to observe if the theory is accurate, enable us to accept, reject or revise the theory

        - single theory → large number of predictions, because 1 hypothesis is usually not sufficient to test the entire theory
        - Most researchers test their theory with a series of empirical studies, each designed to test an individual hypothesis.
    - ***Data*** = set of observations
        - Data may support or challenge a theory
        - If the data does not match the hypothesis, theory needs to be revised
    - ***Weight of the evidence***:
        - How scientists evaluate their theories
        - Weight of the evidence for or against
2. ***Basic research-applied research cycle***
   Basic Research → Translational Research → Applied Research → Back
    - ***Basic Research***
        - = contribute to the general body of knowledge
        - Not intended to address a specific, practical problem
        - not just random gathered facts
        - Solid basic research is an important basis for later applied studies
    - ***Translational Research***
        - = the use of basic research to develop and test better treatment and intervention
        - The dynamic bridge from basic to applied
    - ***Applied Research***
        - = directly applied to the problem in a particular real-world context
        - Can use empirical evidence to answer a nonprofit organisation's questions
    - Example:
        - *Basic Research*: Under what conditions do people experience a rubber hand as their own hand?
        - *Translational Research*: Can lessons from the "rubber hand illusion" be applied to the design of prosthetic hands?
        - Applied Research: Which prosthetic hand design works best for clients?
3. ***Peer-review cycle***
   Editor → Experts → Editor
    - ***Peer-Reviewed Process*** = the editor receives a manuscript → sends to three or four experts on the subject → experts tell the editor about the manuscript's virtues and flaws → editor decides to publish the paper or not
    - Scientific articles are published in peer-reviewed journals
        - Rigorously reviewed by other scientists
        - Anonymous review, so honest assessment
        - Manuscripts can be rejected, or accepted with minor or major revisions
4. ***Journal-to-journalism cycle***
   Scientific Journal → Journalism
    - ***scientific journals:*** are read primarily by other scientists and students
        - the general public almost never reads them
    - ***Journalism***: news and commentary that the general public reads or hears

    - Non-scientists write (and read) about the scientific journals in the “Popular press”
    - Pros:
        - Psychologists can benefit when journalists publicise their research
    - Cons:
        - Sometimes journalist report sensational instead of importance
        - They don’t always get the story right
            - Possibly dumb down the details to make the story more accessible to a general audience
    - Example: ***“Mozart effect”***
        - Idea: Mozart makes you smarter; Classical music makes you smarter when studying
        - Problem: Generalising this to children; Not true

<u>3) The logic behind science</u>
- Conditional argument
- The type of argument used to justify or falsify a scientific hypothesis is the same across different scientific disciplines
    - The conclusion of the argument will be either the hypothesis or its negation

Research
- ***Research*** would require us to administer surveys or experiments
    - Conduct a research study to examine if there is a relationship between frequency of watching violent films and aggressive behaviours.

The scientific method
- ***Operational definition***
    - A statement of procedures (operations) used to define research variables
        - How you are going to measure aggression is your operational definition for aggression
- ***Replication***
    - Repeating the essence of a research study to see whether the result is consistent
    - Usually with different participants in different situations

<u>4) Scientific reasoning</u>
- 1 condition for a test of the hypothesis
    - ***Affirming the antecedent***
      If A then B
      A
      → Thus B
    - ***Denying the consequent***
      If a then B
      Not B
      → Thus not A
- 2 conditions for a good test of the hypothesis
    - If [H and IC and AA] then P
    - If [not-H and IC and AA] then probably not-P

        - H = Hypothesis
        - IC = Initial Conditions (e.g., description of state at beginning of experiment)
        - AA = Auxiliary Assumptions (e.g., registration is reliable)
        - P = Prediction (follows deductively from H and IC and AA)
- Justifying a theoretical hypothesis (inductively)
    - If [not-H and IC and AA] then probably not-P
      P
      Thus: not [not-H and IC and AA]
      H or not-IC or not-AA IC and AA
      → Thus: H
- Refuting a theoretical hypothesis (deductively)
    - If [H and IC and AA] then P
      not-P
      Thus: not [H and IC and AA]
      not-H or not-IC or not-AA
      IC and AA
      → Thus: not-H
- Result → not P
    - If [H and IC and AA] then P
      not-P
      Thus: not [H and IC and AA]
      not-H or not-IC or not-AA
      IC and H
      Thus: not-AA
        - Scientific reasoning This is valid but not scientifically justified!! because the hypothesis HAS TO BE the conclusion

<u>5) Falsifiability criterion</u>
- If we test our ideas, we can know when we are wrong. And if we know when we are wrong, we can correct our mistakes, if we don't, we can't
- Scientific theories should have specific predictions rendering the theory potentially falsifiable

To be a good scientist:
- Create comparison groups and look at all the data
- Generate data through studies
- Don't go with what everyone believes
- Ask questions and collect even disconfirming evidence
- Accept data provisionally and change theories

Features of a good scientific theory:
- Supported by data
    - Scientists need to conduct multiple studies, using a variety of methods to address different aspects of their theories
    - Good theory is supported by a large quantity and variety of evidence
- Are ***falsifiable***/***Tentative***
    - never accepted as absolutely correct

- Are ***Empirical***
    - using evidence from the senses or from instruments that assist the senses as the basis for conclusions
- Are ***parsimonious***
    - the simplest explanation using the fewest possible assumptions
    - Also referred to as “***Occam’s razor***””
- Are ***Rational***
    - They follow the rules of logic and are consistent with known facts
- Are ***Testable***
    - verifiable through observation and can be disproven
    - Contrast with pseudoscience explanations
- Are ***General***
    - apply beyond the original observations on which they are based
- Are ***Rigorously evaluated***
    - constantly evaluated for consistency with evidence and parsimony

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05/10/2023

L10 (RM 3,11)

Validity

<u>1) Basic knowledge</u>

Variables

- Is something that varies, so it must have at least two levels, or “values”
- ***Independent variables***
    - multiple-regression analysis → predictor variable used to explain variance in the criterion variable
    - Experiment → variable that is manipulated.
    - ***Manipulated variables***
        - a variable that a researcher controls/changes
    - E.g., condition - congruence (incongruent; neutral; congruent)
    - ***Conditions***
        - one of the levels of the independent variable in an experiment
- ***Dependent variables*** (outcome variable)
    - multiple-regression analysis → the single outcome, or criterion variable the researchers are most interested in understanding or predicting
    - Experiment → the variable that is measured.
    - ***Measured variables***
        - a variable in a study whose levels (values) are observed and recorded
    - E.g., Time or RT
- ***Confounding variables***
    - Variables that change simultaneously with independent variables
    - E.g., time of testing, order
- ***Control variables***
    - Variables that are being held constant on purpose
    - E.g., number of words, colours etc.
- Random error
    - E.g. noise during testing
- ***Conceptual variable***
    - A variable stated at an abstract level

    - E.g., anxiety, intelligence
    - Sometimes called a "***construct***"
    - ***Conceptual Definitions***:
        - Researcher's definition of a variable at an abstract level/ theoretical level at which conceptual variables must be defined at
- ***Operational variable*** (operational definition)
    - The specific way in which a construct is manipulated or measured in a study
    - Variables should be defined in terms of the operations needed to produce them
    - "How did the researcher manipulate or measure the construct?"
    - E.g., test score → control or anxiety group
    - Operationalizing definitions/ ***To operationalize***:
        - Means to turn a concept of interest into a measured or manipulated variable
- ***Participant variable***
    - a variable such as age, gender, or ethnicity whose levels are selected (i.e. measured), not manipulated

Confederate:
- An actor playing a specific role for the experimenter

Constant
- Is something that could potentially vary but has only one level in the study in question
    - Example: In research on fathers, sex is not a variable

Goals of psychology
1. To describe behaviour and mental processes
2. To predict behaviour and mental processes
3. To explain and understand behaviour and mental processes
4. To influence/control behaviour and mental processes
- To predict behaviours, thoughts, feelings, change and so on…
    - Mainly by way of assessing the relationship between two or more variables

To represent claims
- ***Scatterplot***:
    - A graph in which one variable is plotted on the y-axis and the other variable is plotted on the x-axis
    - Each dot represents one participant in the study, measured on the two variables
    - Types:
        1. Positive
            - High goes with high / high goes with low
        2. Negative
            - High goes with low
            - Low goes with high
        3. Zero
            - No association
        4. Curvilinear

            - One variable *changes* its pattern as the other variable *increases*. (a curve) or vice versa

Three claims (statements)

- = the argument someone is trying to make
- 1) ***Frequency claims***
    - = Describe a rate or level of something or degree of a single variable
    - The variables are always measured, not manipulated
    - Generally focus on one variable
        - ***Descriptive*** research
- 2) ***Association claims***
    - = Argues that one level of a variable is likely to be associated with a particular level of another variable
    - Variables are measured not manipulated
    - Focus on the relationship between at least two variables
    - Making Predictions
        - Correlation does not imply causation
            - ***Third-variable problem***
                - There may be an unmeasured variable that actually causes variables to ***covary*** (change together)
            - ***Directionality problem***
                - Not always possible to specify the direction in which a causal arrow points
            - Don't know what causes what
        - Correlation does allow prediction
            - If you know there is a positive or a negative correlation, you can predict the value of the other variable
            - The stronger the relationship between the variables, the more accurate the prediction
        - Association
            - Correlation - Correlations do not permit inferring causality
            - Covariation
- 3) ***Causal claim***
    - = argue that one variable is responsible for a change in the other
    - Can be positive, negative, or curvilinear
    - Causal claims are *stronger*, and therefore held to a *higher* standard
        - Must show that one came first (causal variable) and other came second (outcome variable)
        - Must establish that no other explanations exist for the relationship
    - Focus on ***causation***:
        - A change in one variable is responsible for changing the value of another variable
        - ***Experimental*** research
        - Factor A → Factor B

How do we know whether these claims are good?

- ***Validity***
    - = The appropriateness of a conclusion or decision of a claim

        - A valid claim is: Reasonable? Accurate? Justifiable?

2) ***Construct Validity***

- = how well a conceptual variable is measured or manipulated (operationalized) in a study
- *How well is a construct operationalized?*
- Threats to construct validity
    - Inadequate operational definition
        - E.g., does a test measure anxiety or absence of self-confidence?
    - Mono-operation bias
        - E.g. mental workload

3) ***External validity***

- An indication of how well the results of a study generalise to, or represent, individuals, settings, places, and times (contexts) besides those in the study itself
- *Is it possible to generalise?*
- Threats to external validity
    - Selection biases
        - E.g., study on the effect of marijuana on the ability to solve maths problems
    - Study setting is different from other settings
        - E.g., survey study on attitudes towards drug users
        - E.g., study on memory in a stuffy classroom
        - E.g., covid spread in urban vs. rural areas
    - Specific time at which the study is performed is different from other times
        - E.g., study on attitude concerning eating meat after BSE scandal

4) ***Internal validity***

- The extent to which it is possible to rule out alternative explanations for a causal relationship between two variables.
- *Can we rule out alternative explanations?*
- Threats to internal validity
    - ***Maturation threat***
        - observed change in an experimental group could have emerged more or less spontaneously over time
    - ***History threat***
        - whether a change in the treatment group is caused by the treatment itself or by an external or historical factor
    - ***Testing threat (practice effect)***
        - scores changed over time just because participants have taken the test more than once
    - ***Instrumentation threat***
        - measuring instrument changes over time
        - E.g., study on the effect of a new educational maths program on maths ability
    - ***Regression to the mean***
        - Expected return to normal

            - E.g., study of a new learning method in low- and medium-ranking pupils
    - ***Attrition threat***/ experimental mortality
        - when a systematic type of participant drops out of the study before it ends
        - E.g., study on the effect of couple counselling on the divorce rate
    - Selection effects
        - ***Selection-history threat***
            - historical or seasonal event systematically affects only the participants in the treatment group or only those in the comparison group, not both
        - ***Selection-attrition threat***
            - participants are likely to drop out of either the treatment group or the comparison group, not both
        - E.g., Simpson paradox
    - ***Design confounds***
        - The presence of another variable that unintentionally varies systematically with the independent variable
        - E.g., study on the effect of bowl-size on estimates calories consumed
    - ***Observer bias***
        - observer expectations influence the interpretation of participant behaviours or the outcome of the study
        - E.g., study on a new antidepressant
        - ***Single- and double-blind studies***
            - Double: study in which neither the participants nor the researchers who evaluate them know who is in the treatment group and who is in the comparison group
            - Single: the observers are unaware of the experimental conditions to which participants have been assigned
    - ***Demand characteristics (experimental demand)***
        - cue that leads participants to guess a study's hypotheses or goals
    - placebo effects
    - ***Order Effects***:
        - in a within-groups design, exposure to one condition changes participant responses to a later condition

5) ***Statistical validity***
- The extent to which a study's statistical conclusions are accurate and reasonable
- *How well do the numbers support the claims?*
- Threats to statistical validity
    - Violated assumptions of the test statistics
    - Fishing and the error rate problem
        - Seeing things that aren't there
    - Low statistical power
        - Missing the needle in the haystack
        - Unreliability of measures

L11 (RM 5,6,7)

Measurement, reliability, and descriptive research

1) Measurement scales

- ***Measurement***
  - A systemic way to assign numbers or names to object or features
- Measurement scales
  - Nominal scales - differences (categorical)
  - Ordinal scale - ranked order
  - Interval scale - equal intervals (add/subtract)
  - Ratio scale - absolute zero (ratio's)
- Importance of measurement scales
  - Determines the possible statistical operations
  - Determines the possible conclusions

2) Reliability

- Observed score = true score + error score
  - ***Observed score***
    - The measured value
  - ***True score***
    - The real value
  - ***Error score***
    - The difference between the measured value and the real value which is affected by
      - The test (situation)
      - The person
- ***Reliability***
  - Consistency of measurements
- Different types of reliability
  - Test reliability
    - Test-retest reliability
    - Internal reliability
      - Split-half reliability
      - Cronbach's alpha
  - Interrater reliability
  - Experimental reliability

3) Descriptive research

- Descriptive methods
  - To *describe* human and animal behaviour and mental processes
  - ***Probabilistic:*** Which means that its findings are not expected to explain all cases all the time
    - Conclusions are meant to explain a certain proportion (very high proportion) of the cases

Descriptive method: ***Observational research***

- The process of watching people or animals and systematically recording how they behave or what they are doing

- To describe nature
    - To generate hypotheses (not to test hypotheses!)
        - E.g., Bystander effect
- Problems
    - People are not always good at observation
        - This is a threat to interrater reliability
    - Low internal validity
    - Threats to construct validity
        - Observer bias
        - Observer effects
        - Reactivity
- To fix it
    - Unobtrusive observation
    - "Wait it out" strategy (participant observation)
    - Unobtrusive measures

Descriptive method: ***Survey research (or poll)***

- A method of posing questions to people on the telephone, in personal interviews, on written questionnaires, or via the internet
    - Question formats
        - Open-ended questions
            - What do you think of this class
        - forced -choice format
            - Do you like this class so far
        - Likert scale
            - I like this class (agree 1-5)
        - Semantic differential format
            - How difficult is this class (1-5)
- Problems
    - Low internal validity
    - Threats to construct validity
        - Question wording
            - Leading question
                - wording encourages one response more than others, thereby weakening its construct validity
            - Double-barreled question
                - asks two questions in one, thereby weakening its construct validity
            - Negatively worded question
                - making its wording complicated or confusing and potentially weakening its construct validity
        - Question order
        - The use of ***response sets***
            - a shortcut respondents may use to answer items in a long survey, rather than responding to the content of each item. Also called ***nondifferentiation***
        - Trying to look good or bad

        - People are not always capable of reporting accurately about their feelings, thoughts, and behaviour

Descriptive method: ***Case studies***

- Descriptive research → frequency claims
- Population & samples
    - ***population*** = the entire set of people or products in which you are interested
    - ***sample*** = a smaller set/subset, taken from that population
    - ***census*** = a set of observations that contains all members of the population of interest
    - Threats to external validity
        - ***Biassed or unrepresentative sample***
            - ***Convenience sampling***:
                - Studies may use samples consisting of only those who are easy to contact
            - ***Self-selection***:
                - Many studies use samples consisting of only those who volunteer to participate
        - To obtain a representative sample there are various techniques
            - Random sample
            - Different rewards
            - Replication
    - Threats to statistical validity
        - A too small sample size
    - The internal validity of frequency claims is low
    - It is important to question how well
        - Each variable was measured (construct validity)
        - The results generalise (external validity)
        - The numbers support the conclusions (statistical validity)
        - Reliability

12/10/2023

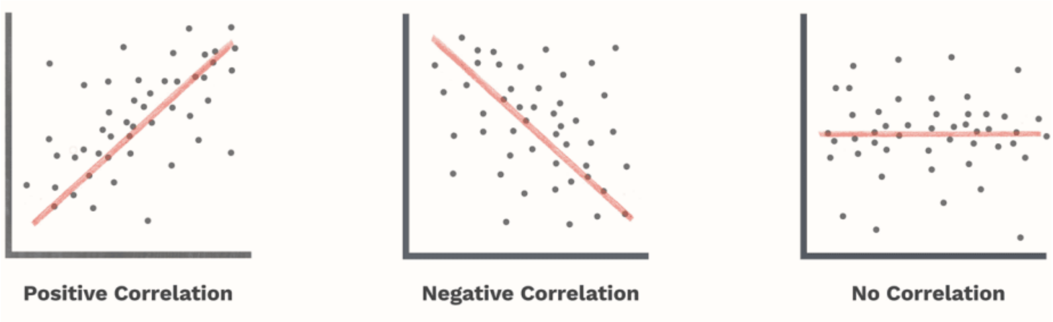
L12 (8,9,10,11)

Correlations and Experiments

<u>1) Correlational research</u>

- Correlational research → association claims
    - "People who often watch violence on TV are more violent than others"
    - "Mental disorders are more common among unemployed people than among people with a job"
    - "Couples who meet online have better marriages"
    - "In countries with a high number of divorces, people generally have a high life expectancy"
- ***Correlation coefficient*** (r)
    - A measure indicating the degree of linear relation between two variables
        - Quantify the relationship
    - Varies between -1.00 and +1.00
        - Absolute value indicates the *strength* of the relation
        - Sign indicates the *direction* of the relation

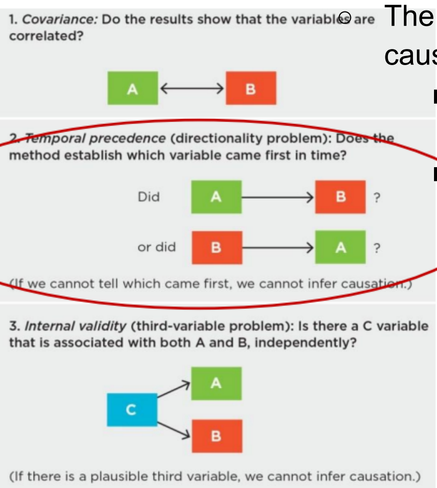
    - Scatter diagram



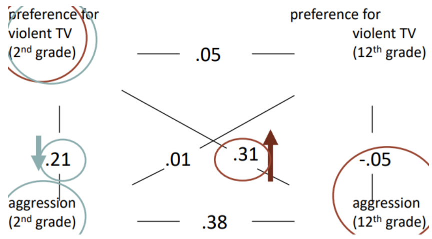
        - Also: ***bivariate correlation*** = an association that involves exactly two variables
- Correlational research → association claims
    - Statistical validity: do the data support the claim?
        - What is the strength of the correlation (***effect size***)?
            - the magnitude, or strength, or a relationship between two or more variables
        - Could outliers be affecting the correlation?
        - Is the correlation ***statistically significant***?
            - unlikely the result came from the null-hypothesis population
- There exists a positive correlation between
    - Hours people do sports per week
    - General health

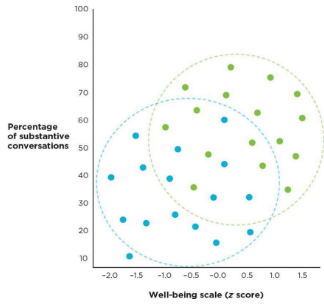
      v
    - What does this imply?
        - Sports → health
        - Health → sports
    - Unknown direction of causation (directionality problem)
- There exist a positive correlation between
    - The number of ice creams sold per month
    - The number of people drowned per month

      v
    - What does this imply?
        - Ice creams → drowning
        - Drowning → ice creams
        - High temperature → more ice creams and drowning
    - 3rd variable problem
- Correlational research → association claims



    - The presence of a correlation indicates an association but does not indicate causality!
        - ***Directionality problem*** = sometimes researchers attempt to establish temporal precedence based on correlational research.
        - both variables being measured around the same time, making it unclear which variable in the association came first.
            - → How? By using the cross-lagged-panel procedure
                - $A \rightarrow B$ if $r(A(t1), B(t2)) > r(A(t1), B(t1))$

○ 



- ■ ***3rd variable problem*** = Sometimes researchers attempt to establish (some) internal validity based on correlational research.
- ■ existence of a plausible alternative explanation for the association between two variables
  - → How? By separating the participants into subgroups on the basis of a reasonable 3rd variable

○ What if the correlation is zero?
- ■ ***Spurious association:***
  - only systematic mean differences on subgroups within the sample; the original association is not present
- ■ ***Restriction of range*** (or truncated range)
  - in a bivariate correlation, the absence of a full range of possible scores on one of the variables
- ■ Non-linear relationship
- ■ No association

○ The internal validity of associations claims is low

○ It is important to question how well
- ■ Each variable was measured (construct validity)
- ■ The results generalise (external validity)
- ■ The numbers support the conclusions (statistical validity)

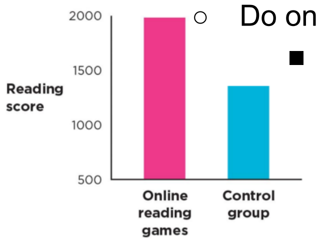
2) The basics of experimenting

Experiment groups

- Enables us to compare what would happen both with and without the thing we are interested in
- ***Comparison Group*** = a group whose levels on the independent variable differ from those of the treatment group in some intended and meaningful way.
- ***Control Group*** = a level of an independent variable that is intended to represent “no treatment” or a neutral condition.
- ***Treatment Group*** = the participants in an experiment who are exposed to the level of the independent variable that involves a medication, therapy, or intervention
- ***Placebo (control) Group*** = a control group in an experiment that is exposed to an inert treatment such as a sugar pill.
- ***Matched Groups*** = an experimental design or technique in which participants who are similar on some measured variable are grouped into sets; the members of each matched set are then randomly assigned to different experimental conditions.
- Four cell columns:

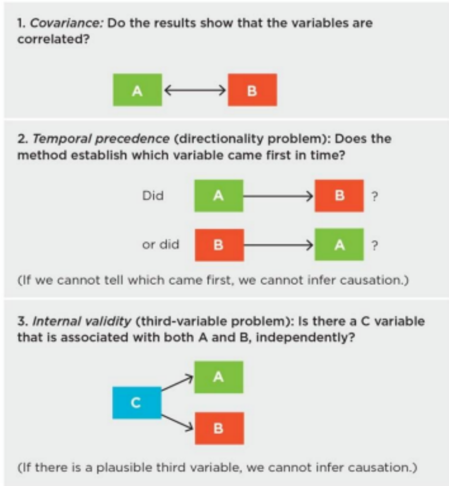
- Untreated/improved cell
- Treated/improved cell
- Untreated/unimproved cell
- Treated/unimproved cell

A hypothetical experiment



- Do online reading games make kids better readers?
    - independent variable:
        - Reading training
            - Online reading games (experimental group)
            - No online reading games (control group)
    - dependent variable:
        - Reading ability
            - Operationalized in terms of score on reading test

Note: these results are fabricated!



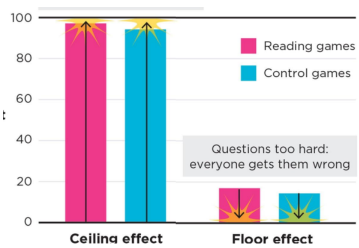
3) Experimental research

Experiment designs

- ***Multivariate designs*** = a study designed to test an association involving more than two measured variables
- ***Longitudinal design*** = a study in which the same variables are measured in the same people at different points in time
    - Correlations in longitudinal design:
        - ***Cross-sectional correlation*** = correlation between two variables that are measured at the same time
        - ***Autocorrelation*** =correlation of one variable with itself, measured at two different times
        - ***Cross-lag correlation*** =correlation between an earlier measure of one variables and a later measure of another variable
- For independent variable:
    - ***Between-subjects design*** (***independent-groups design***)
        - Separate groups of subjects receive different levels of the independent variable(s), such that each participant experiences only one level of the independent variable
        - ***Posttest-Only Design*** = using an independent-groups design, participants are tested on the dependent variable only once.
        - ***Pretest/Posttest Design*** = using an independent-groups design, participants are tested on their key dependent variable before and after exposure to the independent variable.
    - ***Within-subjects design*** (***within-groups/repeated-measures design***)
        - All subjects receive all levels of the independent variable(s).
        - ***Repeated Measures Design*** = an experiment using a within-groups design in which participants responded to a dependent variable more than once, after exposure to each level of the independent variable
        - ***Concurrent-Measures Design***: an experiment using a within-groups design in which participants are exposed to all levels of an

independent variable at roughly the same time, and a single attitudinal or behavioural preference is the dependent variable
- ***Manipulation Check*** = in an experiment, an extra dependent variable researchers can include to determine how well a manipulation worked
- ***Pilot Study*** = a study completed before (or sometimes after) the study, usually to test the effectiveness or characteristics or the manipulations

Null results (Null effect)(H0)
- The independent variable did not affect the dependent variable; there is no significant covariance between the two
- Manipulation was too weak
  - E.g., 1 week of training might have been too short
- Insensitive measures
  - E.g., tiny improvements in reading ability might not have been detectable in a simple reading test
- ***Ceiling effect***
  - all scores fall at the high end of their possible distribution
  - E.g., reading test might have been too easy
- ***Floor effect***
  - all scores fall at the low end of their possible distribution
  - E.g., the reading test might have been too difficult.



- Unreliable measurements
  - ***Noise*** (error variance, unsystematic variance)
    - unsystematic variability among the members of a group in an experiment
    - results from: situation noise, individual differences, measurement error
      - ***Situational Noise*** = unrelated events or distractions in the external environment
      - ***measurement error*** = the degree to which the recorded measure for a participant on some variable differs from the true value of the variable for the participant
- Not enough power (small n)

16/10/2023

L13 (RM 10,12)

Single-factor and factorial designs

<u>1) Single-factor designs</u>

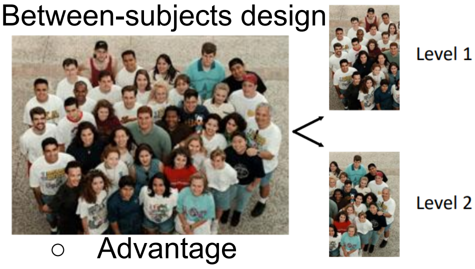
Single-factor designs
- How to assign subjects to the different levels of the independent variable(s)?
  - Between-subjects design (independent-groups design)
    - Separate groups of subjects to receive different levels of the independent variable(s).
  - Within-subjects design (within-groups / repeated-measures design)
    - All subjects receive all levels of the independent variable(s).

Within & Between Subjects Designs
- Independent variable
  - Dependent variable

- Location of lesion in the cortex
    - % correct
- Rotation speed of radar arm
    - % detected
- Learning method
    - Performance on test
- Odour
    - Score concerning the attractiveness of person

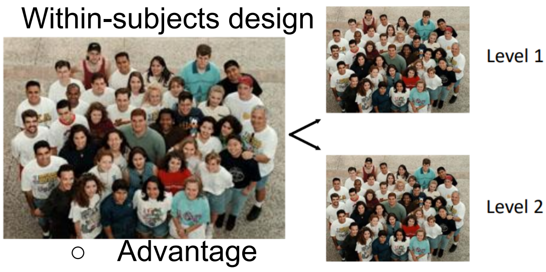
Single-factor designs

- Between-subjects design



← independent variable

    - Advantage
        - There is no chance that one level of the independent variable affects performance under the other level of the independent variable.
    - Disadvantage
        - There is a chance that groups differ.
            - Therefore: random assignment
            - Therefore: matching (matched groups)
    - Use when
        - The independent variable has a permanent effect.
        - There is a large risk of order effects
            - ***Practice effect*** = participants' performance improves over time because they become practised at the dependent measure
            - ***Carryover effect*** = some form of contamination carries over from one condition to the next
- Within-subjects design



← independent variable

    - Advantage
        - There is no chance of group differences:
            - Each subject serves as their own control.
    - Disadvantage
        - There is a chance of order effects (practice, carryover)
            - Therefore
                - ***Counterbalancing*** = presenting the levels of the independent variable to participants in different sequences to control for order effects
                    - Complete counterbalanced design (***full counterbalancing***) = all possible condition orders are represented

        - ***Partial Counterbalancing*** = a method of counterbalancing in which some of the possible condition orders are represented
    1. ***Latin square design***
        - every condition in a within-groups design appears in each position at least once
    2. Balanced latin square design
    3. Random order
        - 1+2+3 → Partial counterbalancing
    - Use when
        - The independent variable does not have a permanent effect.
        - Order effects (practice, carryover) can be avoided by counterbalancing or random order.
- Within-subject designs are more efficient than between-subjects designs (more power).

2) Factorial designs

- Design in which each level of every independent variable occurs with all levels of the other independent variable
    - Example:
        - Blood Alcohol Content (BAC) with three levels: 0‰, 0.5‰, 1.0‰ (per mil)
        - Power steering with two levels: with power steering, without power steering

Power Steering	**Blood Alcohol Content**		
	0	0.5	1.0
without	without, 0	without, 0.5	without, 1.0
with	with, 0	with, 0.5	with, 1.0

- Factorial combinations of all levels of BAC (3) and Power Steering (2)
- This results in 3 x 2 = 6 conditions (number of ***cells*** in table).
    - one of the possible combinations of two independent variables
- BAC and Power Steering vary independently of each other.
- They are not confounded

General format

- Letters: factors (independent variables) - A, B, C, …
- Figures: level of factor - 1, 2, 3, …
- Example of a "3 x 2 factorial design"
    -

	A1	A2	A3
B1	A1B1	A2B1	A3B1
B2	A1B2	A2B2	A3B2

        - There are 3 x 2 = 6 conditions
- Example of a "2 x 2 x 2 factorial design"
    -

C2

	A1	A2
B1	A1B1C2	A2B1C2
B2	A1B2C2	A2B2C2

        - Three factors (A, B, C), all with two levels (1, 2)
        - Eight conditions

Different designs (for example 1)
- ***Between-subjects design***: A and B are both between-subjects factors. Six groups.
- ***Within-subjects design***: A and B are both within-subjects factors, e.g., six sequences (in Latin Square)
- ***Mixed design***: one of the factors is a between-subjects factor; the other factor is a within-subjects factor.

Advantages factorial design
- ***Efficiency***
    - E.g., one experiment with two factors instead of two experiments, each with one factor.
- ***Generalisation***
    - E.g., the effect of BAC occurs both in cars with and without power steering
- ***Information on interaction between factors***
    - E.g., does the effect of BAC depend on the presence of power steering

Effects in a two-factorial design
- Main effect of Factor A (BAC)
- Main effect of Factor B (Power steering)
- Interaction effect A x B
    - ***Main effect***: The overall effect of one independent variable on the dependent variable, averaged over the levels of the other independent variable(s)
    - ***Interaction effect:*** the difference in the levels of one independent variable changes, depending on the level of the other independent variable; a difference in differences
        - Symmetrical: interaction A x B = interaction B x A
    - ***Marginal means***: the arithmetic means for each level of an independent variable, averaging over the levels of another independent variable

<u>3) Experimental research</u>
- Experimental research → causal claims
    - Experimental research establishes internal validity
    - However, this requires appropriate designs!
        - Were there any design confounds?
            - ***Between-subjects variables***:
                - Did the researcher control for selection effects using random assignment or matching
            - ***Within-subjects variables***:
                - Did the researcher control for order effects by (partial) counterbalancing or randomization?
    - It is also important to question how well
        - Each variable was manipulated and measured (construct validity)
        - The results generalise (external validity)
        - The numbers support the conclusions (statistical validity)

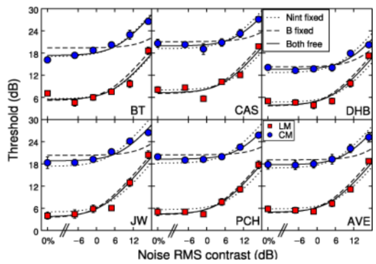
19/10/2023

L14 (RM 13)

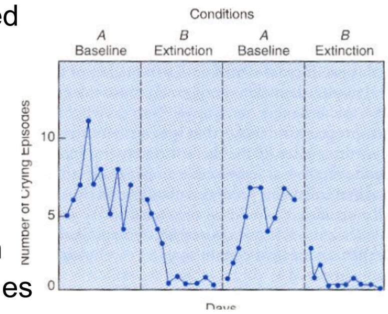
Small-N and quasi experimental designs

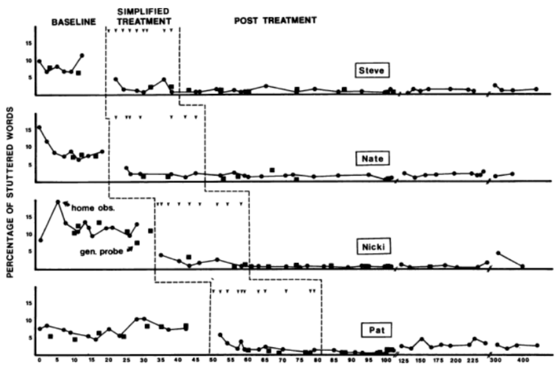
1) <u>Small-N studies</u>

- = a study in which researchers gather information from just a few cases



- Psychophysics
  - Psychophysical study
  - N = 5
  - Participants: BT, CAS, DHB, JW, PCH
  - AVE = average
  - *"If you've seen one, you've seen them all"*
- Behaviour-change studies
  - ***Stable-baseline design*** (AB Design)
    - = researcher observes behaviour for an extended baseline period before beginning, and continues observing behaviour after the intervention
    - A = behaviour during baseline
    - B = behaviour during treatment / after treatment



  - ***Reversal design*** (ABA or ABAB Design)
    - = researcher observes a problem behaviour both before and during treatment, and then discontinues the treatment for a while to see of the problem behaviour returns
  - ***Multiple-baseline design***
    - = a small-N design in which researchers stagger their introduction of an intervention across a variety of contexts, times, or situations
    - 

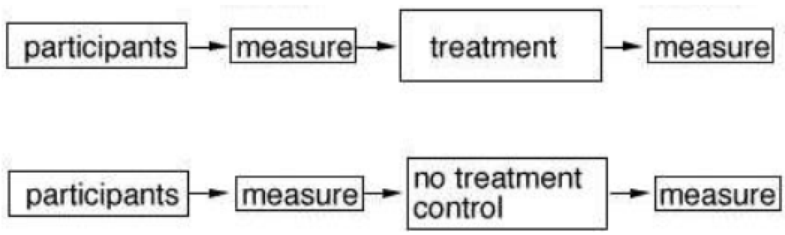
- ***Single-N Design*** = a study in which researchers gather information from only one animal or one person
- Case studies

- Small-N studies → often aim to generate causal claims
  - Internal validity is often not high
  - It is also important to question how well
    - each variable was manipulated and measured (construct validity)
    - the results generalise (external validity)
    - the numbers support the conclusions (statistical validity)

2) <u>Quasi-experimental designs</u>

- ***Quasi-Experiments*** = a study similar to an experiment except that researches do not have full experiment control (e.g., they may not be able to randomly assign participants to the independent variable conditions)
    - Experiments in which the experimenter selects rather than manipulates the independent variables.
- ***Quasi-Independent Variable*** = a variable that resembles an independent variable, but the researcher does not have true control over it (e.g., cannot randomly assign participants to its levels or cannot control its timing)
    - Environmental variables
    - Subject variables
- Pretest/posttest design (AB design)
  O1 - T - O2
    - E.g., "study on the effectiveness of flower essence therapy on depression"
- ***Nonequivalent Control Group Posttest-Only Design***
    - = participants in one group are exposed to a treatment, a nonequivalent group is not exposed to the treatment, and then the two groups are compared.
- ***Nonequivalent control group pretest/posttest design***
    - = a quasi-experiment that has at least one treatment group and one comparison group, in which participants have not been randomly assigned to the two groups, and in which at least one pretest and one posttest are administered.

O1 - T - O2 experimental group (E)
O1 - - O2 nonequivalent control group (C)



- Matching →
- E.g., "Is there a change in productivity after the introduction of a flex working schedule?
    - E = branch in Amsterdam
    - C = branch in Rotterdam
- ***Interrupted time-series design***
    - = a quasi-experiment in which participants are measured repeatedly on a dependent variable before, during, and after the "interruption" caused by some event
- O1 O2 O3 O4 O5 T O6 O7 O8 O9 O10
    - E.g., "Effect of fluoride in drinking water on school performance."
- ***Nonequivalent control group interrupted time-series design***
    - = a quasi-experiment with two or more groups in which participants have not been randomly assigned to groups; participants are measured repeatedly on a dependent variable before, during, and after the "interruption" caused by some event, and the presence or timing of the interrupting event differs among the groups

O1 O2 O3 O4 O5 T O6 O7 O8 O9 O10
O1 O2 O3 O4 O5 O6 O7 O8 O9 O10

    - E.g., “Effect replacement of crossing by roundabout”
- The subject variable “Age”
- Quasi-experimental studies → aim to generate causal claims
    - Internal validity is not as high as in a true experiment
    - It is also import to question how well
        - each variable was manipulated and measured (construct validity)
        - the results generalise (external validity)
        - the numbers support the conclusions (statistical validity)

<u>3) Questions</u>
A politician believes that the use of car lights during the day contributes to greater safety on the road in the Netherlands.
He relies on a literature review written by a major transportation company. The review reports several studies showing that the number of road accidents decreased significantly in Sweden after the use of car lights during the day became mandatory.
The conclusion of the review was: Based on the results of previous studies (i.e., those performed in Sweden), the use of car lights during the day in the Netherlands should lead to a higher safety on the road.
Why is this conclusion not valid?

A researcher wants to know whether the volume of background noise has an effect on the speed with which people solve calculation assignments.
They perform an experiment in which they manipulate the following independent variable:
(A) Volume of background noise (0 dB, 40 dB, and 80 dB).
A constitutes a within-subjects variable.
Six participants take part in the study.
He uses a complete counterbalanced design.
Indicate the order of the conditions for the different participants.

Chess Grandmaster Alexander Aljechin was convinced that the presence of his cat had a positive effect on his chances to win a chess game.
As a result of this belief, he took his cat to each game he had to play.
Suppose that Aljechin took his cat to a chess tournament, and he won all games.
Why would this outcome not provide sufficient justification for the hypothesis that his cat is a win-inducing organism?

An owner of a "health shop" claims that the daily intake of extra vitamins leads to a better health condition.
He relies on a large study carried out by another health shop owner.
This person recruited subjects for his study in the waiting room of a GP by asking them if they were willing to take extra vitamin pills for a period of 4 months.
All patients complied and used extra vitamin pills for four months.
Fifty subjects took part in the experiment.
Immediately after they were recruited, all participants took part in a test to determine how much time it took them to walk a distance of 1 km.
After four months (and swallowing many extra vitamins) participants took the same test again.
The results:

-average time of the 1st measurement (jan): 10 min
-average time of the 2nd measurement (jun): 7 min
People needed less time to walk 1 km after compared to before the treatment. Based on these results it was concluded that extra vitamin intake has a positive effect on health conditions.
How is this conclusion not valid?
How can we fix the design?

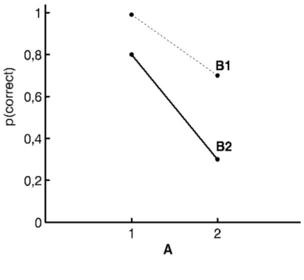
A study includes the following independent variables:
(A) sex (male vs. female) and
(B) use of anabolic steroids (placebo versus anabolic steroids).
The dependent variable is the number of times someone is capable of jumping over a fence (for 2 minutes).
The results (averages of 50 subjects):

- male placebo	5.5
- male anabolic steroids	10.5
- female placebo	0.5
- female anabolic steroids	3.5

The conclusion is threefold:
1) Male participants are better than female participants in jumping.
2) Anabolic steroids improve jumping performance.
3) The effect of anabolic steroids is larger in male participants than in female participants.
Which of these three conclusions is/are not valid and why not?
How can this study be improved?

In a hypothetical study it was shown that there is a positive correlation between the number of times people go to a bar and the amount of alcohol consumption.
A scientist is interested in the question whether going to a bar leads to the consumption of more alcohol or the consumption of more alcohol leads to more bar visits.
Indicate how the scientist might investigate this.
Which result would be in line with the conclusion that going to a bar leads to higher alcohol consumption?

A survey company has just released a new test which was developed to determine someone's decision capability.
The 2-hour test consisting of 50 items should enable companies to select candidates for high-profile jobs.
A critical researcher likes to know how reliable the test is.
To determine this, he submits 50 participants to the test and calculates the correlation between the scores obtained in one half of the test and those obtained in the other half of the test.
The correlation between the two halves of the test is +.05
What can you conclude about the reliability and the validity of the test?



The graph shows the results of a detection experiment.
The statistical analyses show amongst other a significant interaction between A and B.
This implies that the effect of A is different for B1 than for B2.

Is this conclusion valid?
If not, how can the experiment be improved?